JOHARI MBBS ANATOMY
VOLUME 1



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Dr. Vishal Johari is a favourite Youtube figure in the field of Anatomy and make learning intersting by 3D Anatomy model and lectures videos (350+ lecture videos) He is Honoure by Teacher Excellence 2022 Award By Dainik Bhaskar and Eminence Award 2023 by Ministry of Youth and Affairs

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11) PROSTRATE GLAND , UTERUS
15) THYROID , CEREBELLUM
16) TESTIS, OVARY
17) LARGE ARTERY & LARGE VEIN





UPPER LIMB MUSCLES

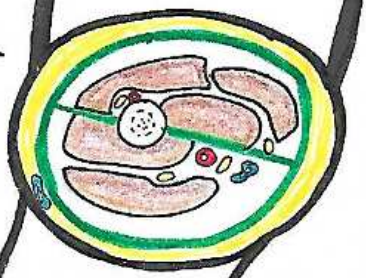
- Radial Nerve
- Radial Nerve
- Posterior interosseous Nerve

- Musculocutaneous Nerve
- Median Nerve
- Ulnar Nerve

ARM

POSTERIOR COMPARTMENT
- TRICEPS BRACHI

ANTERIOR COMPARTMENT
- BICEP BRACHII
- BRACHIALIS
- CORACOBRACHIALIS



FOREARM

POSTERIOR / EXTENSOR COMPARTMENT

SUPERFICIAL
- ANCONEUS
- BRACHIORADIALIS
- EXTENSOR CARPI RADIALIS LONGUS
- EXTENSOR CARPI RADIALIS BREVIS
- EXTENSOR DIGITORUM
- EXTENSOR DIGITI MINIMI
- EXTENSOR CARPI ULNARIS

DEEP
- SUPINATOR
- ABDUCTOR POLLICIS LONGUS
- EXTENSOR POLLICIS BREVIS
- EXTENSOR POLLICIS LONGUS
- EXTENSOR INDICIS

ANTERIOR / FLEXOR COMPARTMENT

SUPERFICIAL
- PRONATOR TERES
- FLEXOR CARPI RADIALIS
- PALMARIS LONGUS
- FLEXOR DIGITORUM SUPERFICIALIS
- FLEXOR CARPI ULNARIS

DEEP
- FLEXOR DIGITORUM PROFUNDA
  - MEDIAL 1/2
  - LATERAL 1/2
- FLEXOR POLLICIS LONGUS
- PRONATOR QUADRATUS



HAND

- THENAR
  - Abductor Pollicis brevis
  - Flexor pollicis brevis
  - Opponens pollicis
  - Adductor pollicis
- HYPOTHENAR
  - Abductor digiti minimi
  - Flexor digiti minimi
  - Opponens digiti minimi
- LUMBRICALS
  - 1st & 2nd
  - 3rd & 4th
- INTEROSSEI
  - Palmar
  - Dorsal

- Median Nerve
- Deep branch of Ulnar Nerve

BRACHIAL PLEXUS

- BRACHIAL PLEXUS IS A NETWORK OF NERVE WHICH SUPPLY UPPER LIMB
- CONSIST ⟶ ROOTS, TRUNKS, DIVISIONS, CORDS AND BRANCHES.
- ROOTS ⟶ CONSTITUTED BY ANTERIOR PRIMARY RAMI C5 – C8 AND T1.
  - ORIGIN OF PLEXUS MAY SHIFT UPWARD & DOWNWARD

PREFIXED PLEXUS ⟶ C4 IS LARGE AND T2 ABSENT

POSTFIXED PLEXUS ⟶ T1 LARGE, T2 PRESENT, C4 ABSENT C5 REDUCED IN SIZE

- TRUNK ⟶ C5 & C6 – UPPER TRUNK
  C7 – MIDDLE TRUNK
  C8 & T1 – LOWER TRUNK

- DIVISION ⟶ EACH TRUNK DIVIDE INTO VENTRAL & DORSAL DIVISION

- CORD ⟶ LATERAL CORD – UNION VENTRAL DIVISION OF UPPER & MIDDLE TRUNK
  MEDIAL CORD – VENTRAL DIVISION OF LOWER TRUNK
  POSTERIOR CORD – DORSAL DIVISION OF ALL 3 TRUNK

- BRANCHES ⟶ BRANCH FROM ROOT – DORSAL SCAPULAR NERVE
  – LONG THORACIC NERVE

BRANCH FROM TRUNK – SUPRASCAPULAR NERVE
– NERVE TO SUBCLAVIUS

LATERAL CORD – LATERAL PECTORAL NERVE (C5-C7)
– LATERAL ROOT OF MEDIAN (C5-C7)
– MUSCULOCUTANEOUS NERVE (C5-C7)

MEDIAL CORD – MEDIAL PECTORAL (C8 , T1)
– MEDIAL CUTANEOUS NERVE OF ARM (C8,T1)
– MEDIAL CUTANEOUS NERVE OF FOREARM (C8,T1)
– ULNAR (C7,C8,T1)
– MEDIAL ROOT OF MEDIAN NERVE (C8-T1)

POSTERIOR CORD – UPPER SUBSCAPULAR (C5 , C6)
– LOWER SUBSCAPULAR NERVE (C5-C6)
– NERVE TO LATISSIMUS DORSI (C6-C8)
– AXILLARY NERVE (C5-C6)
– RADIAL NERVE

HOW TO REMEMBER

LATERAL CORD : LATA LOVE MANI

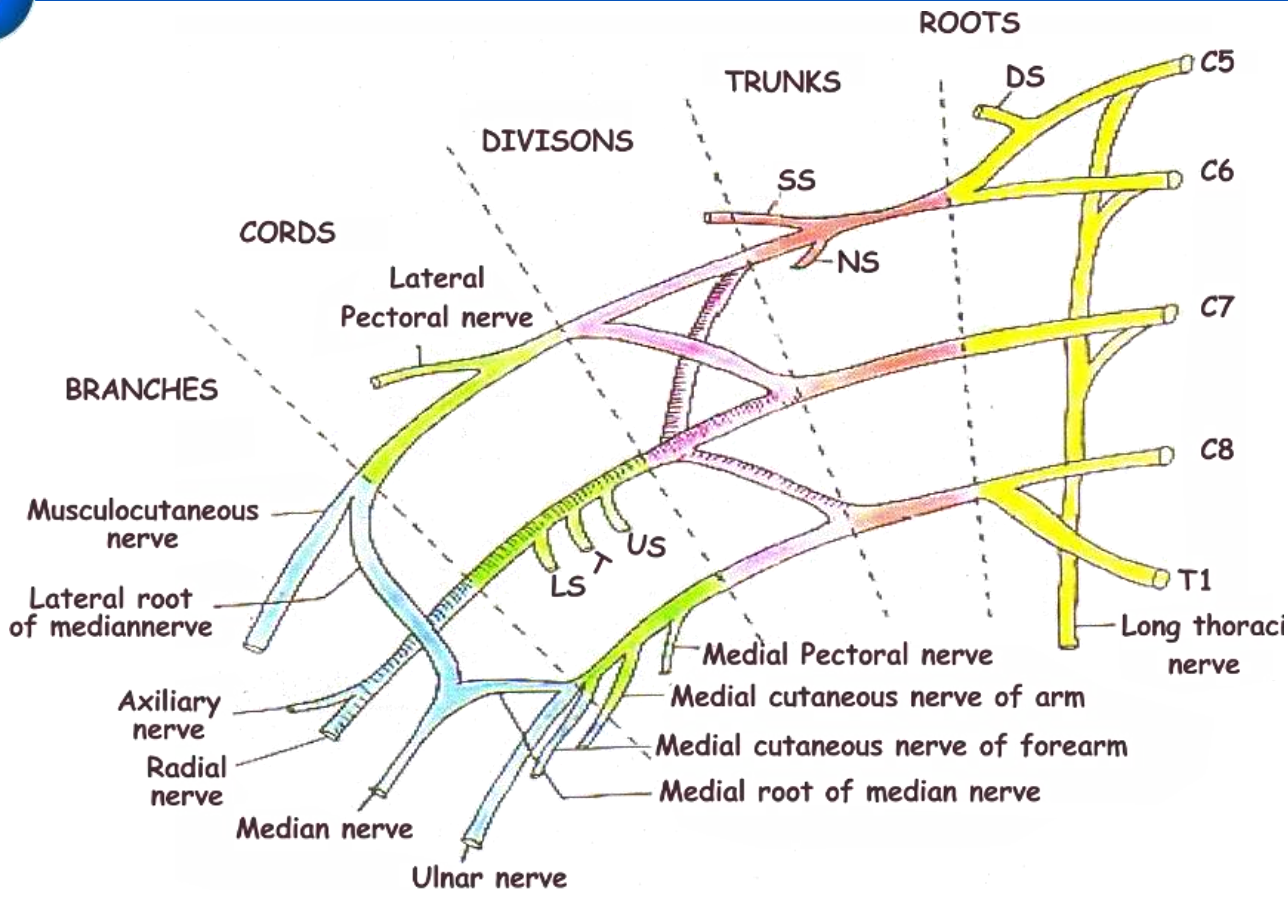
MEDIAN CORD : MOST MEDIAL MEN USE MORPHINE

POSTERIOR CORD : U L N A R

IMPORTANT INFORMATION

SUPRASCAPULAR NERVE –

- THE SUPRASCAPULAR NERVE USUSALLY ARISE AS THE FIRST BRANCH OF UPPER TRUNK BUT IT FREQUENTL SPRING DIRECTLY FROM THE VENTRAL PRIMARY RAMUS OF C5
- IT INNERVATE SUPRSPINATUS AND INFRASPINATUS



Brachial plexus and its branches

[SS - Suprascapular nerve
US - Upper Subscapular nerve
T - Thoraco dorsal nerve
NS - Nerve to subclaviuo
LS - Lower subscapular nerve
DS - Dorsal scapular nerve]

CLINICAL ANATOMY

ERB'S PARALYSIS

SITE OF INJURY - ERBS POINT IN UPPER TRUNK , SIX NERVE MEET HERE

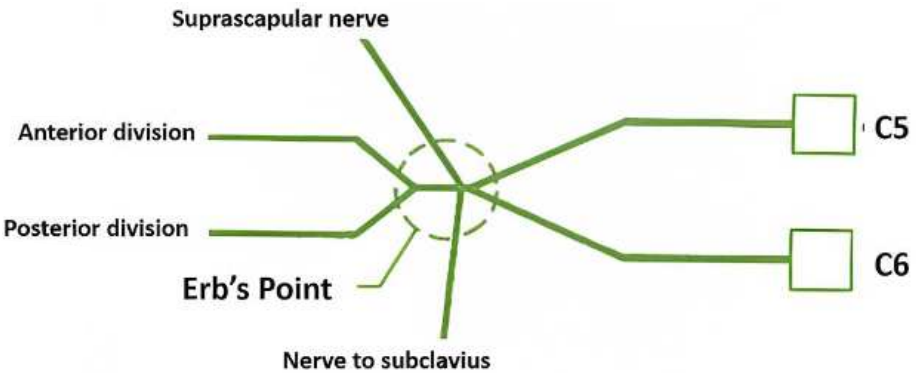
CAUSE OF INJURY - BIRTH INJURY, FALL ON SHOULDER, DURING ANASTHESIA

NERVE ROOT INVOLVE - MAINLY C5 AND PARTLY C6

MUSCLE PARALYSED - MAINLY BICEP BRACHII, DELTOID, BRACHIALIS AND BRACHIORADIALIS
- PARTLY SUPRASPINATUS, INFRASPINATUS AND SUPINATOR

DEFORMITY - ARM - ADDUCTED AND MEDIALLY ROTATED
FOREARM - EXTENDED & PRONATED
THE DEFORMITY ALSO KNOWN AS POLICEMAN'S TIP HAND

DISABILITY - MOVEMENTS ARE LOST - ABDUCTION & LATERAL ROTATION OF ARM & SHOULDER
- FLEXION & SUPINATION OF FOREARM
- BICEPS AND SUPINATOR JERK ARE LOST



KLUMPKE'S PARALYSIS

SITE OF INJURY - LOWER TRUNK OF BRACHIAL PLEXUS
NERVE ROOT INVOLVE - MAINLY T1 & PARTIALLY C8

MUSCLE PARALYSED - INTRINIC MUSCLE OF HAND (T1)
- ULNAR FLEXOR OF WRIST AND FINGER (C8)

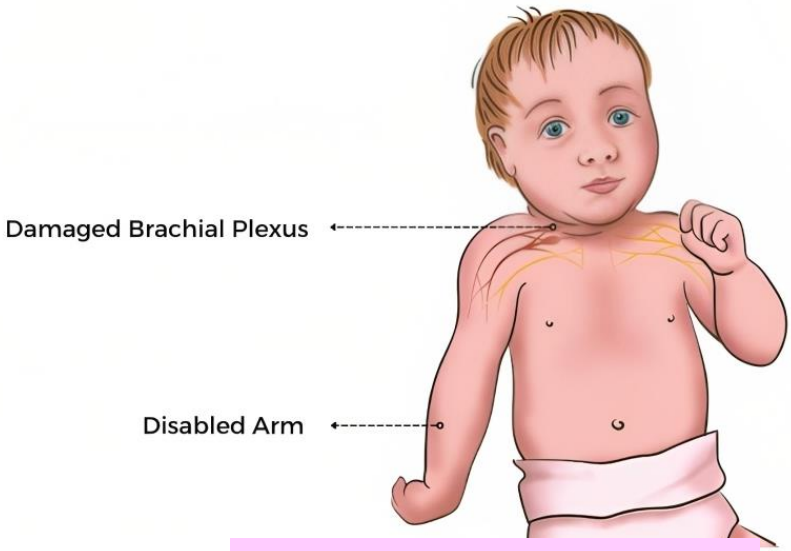
DEFORMITY - CLAW HAND - DUE TO UNOPPOSED ACTION OF LONG FLEXORS AND EXTENSORS
CLAW HAND - HYPEREXTENSION AT METACARPOPHALANGEAL JOINT
-FLEXION AT INTERPHALANGEAL JOINT

DISABILITY - COMPLETE CLAW HAND
- CUTANEOUS ANASTHESIA IN NARROW ZONE ALONG ULNAR BORDER OF FOREARM & HAND

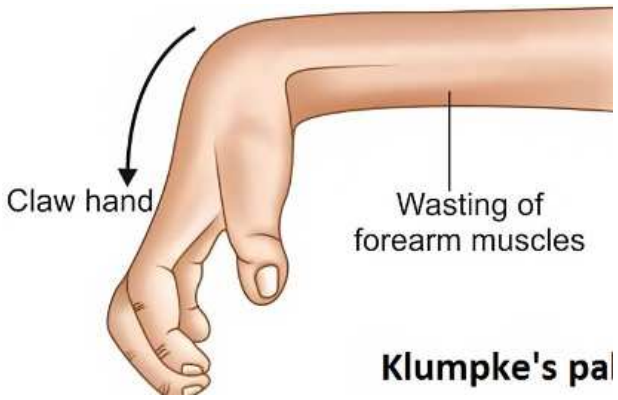
HORNER SYNDROME - INJURY TO SYMPATHETIC FIBERS TO HEAD AND NECK THAT LEAVE THE SPINAL CORD THOUGH NERVE T1
- THERE IS -> PTOSIS, MIOSIS, ANHYDROSIS, ENOPHTHALMOS AND LOSS OF CILIOPINAL REFLEX

WINGING OF SCAPULA

CAUSE - SUDDEN PRESSURE ON SHOULFER FROM ABOVE & CARRY HEAVY LOADS ON SHOULDER
DEFORMITY - WINGING OF SCAPULA - EXCESSIVE PROMINENECE OF MEDIAL BORDER OF SCAPULA
DISABILITY - LOSS OF PUSHING AND PUNCHING ACTIONS
- OVERHEAD ABDUCTION OF SHOULDER GIRLDLE IS PARTLY AFFECTED DUE TO INACT TRAPEZIUS MUSCLE



ERB'S PARALYSIS



KLUMPLE'S PARALYSIS



WINGING SCAPULA

MUSCULOCUTANEOUS NERVE

MUSCULOCUTANEOUS NERVE IS THE NERVE OF FRONT OF THE ARMS

ORIGIN - ARISE FROM THE LATERAL CORD OF BRACHIAL PLEXUS IN AXILLA (C5, C6, C7)

COURSE - RUN DOWNWARD AND LATERALLY --→ PIERCE CORACOBRACHIALIS & SUPPLIES
- THEN PASS BETWEEN BICEPS & BRACHIALIS MUSCLE
- IT APPEARS AT LATERAL MARGIN OF BICEPS TENDON → PIERCE THE DEEP FASCIA JUST ABOVE THE ELBOW
- & DESCEND OVER THE LATERAL ASPECT OF FOREARM → LATERAL CUTANEOUS NERVE OF FOREARM

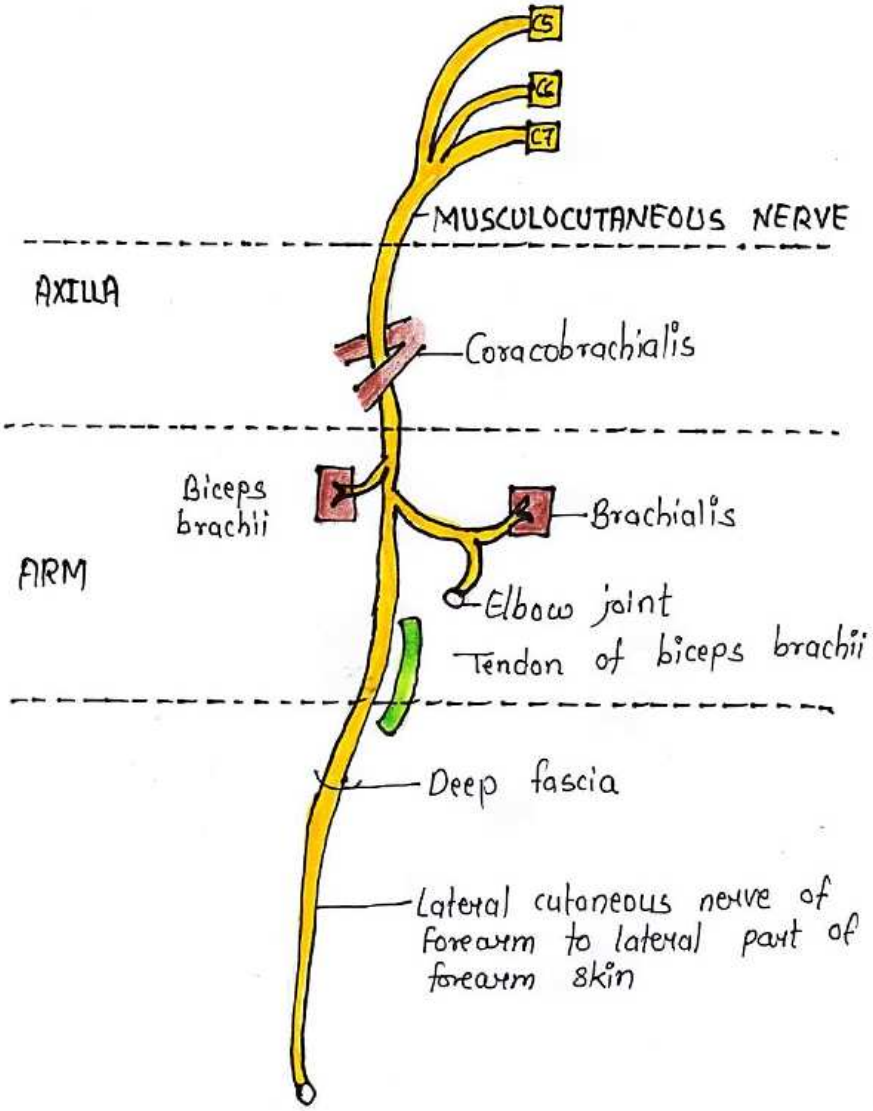
BRANCHES & DISTRIBUTION

1) MUSCULAR BRACNCHES - BICEP BRACHII, CORACOBRACHIALIS & BRACHIALIS
2) CUTANEOUS BRANCH - LATERAL CUTANEOUS NERVE OF FOREARM
3) ARTICULAR BRANCH - TO ELBLOW JOINT THROUGH ITS BRANCH TO BRACHILAIS MUSCLE

CLINICAL ANATOMY

INJURY TO MUSCULOUTANEOUS NERVE

- RARE - LEAD TO FOLLOWING SIGN & SYMPTOMS : LOSS OF STRONG FLEXION & SUPINATION
: LOSS OF BICEPS TENDON REFLEX
: LOSS OF SENSATION ALONG LATERAL ASPECT OF FOREARM



MEDIAN NERVE

MEDIAN NERVE IS THE MAIN NERVE OF FOREARM (C5,6,7,8 & T1)

ORIGIN – ARISE FROM TWO ROOT – LATERAL & MEDIAL ROOT
LATERAL ROOT (C5,6,7) – ARISE FROM LATERAL CORD OF BRACHIAL PLEXUS
MEDIAL ROOT (C8 & T1) – ARISE FROM MEDIAL CORD OF BRACHIAL PLEXUS

COURSE –

- AXILLA – LIES – LATERAL SIDE OF THIRD PART OF AXILLARY ARTERY
  – ENTER ARM AT LOWER BORDER OF TERES MAJOR
- ARM – MEDIAL NERVE LIE LATERAL TO BRACHIAL ARTERY
  - THEN CROSS IN FRONT OF ARTERY FROM LATERAL, TO MEDIAL SIDE
  - THEN ENTER CUBITAL FOSSA.
- CUBITAL FOSSA – LIE MEDIAL TO BRACHIAL ARTERY & TENDON OF BICEPS BRACHII
  -THEN LEAVE FOSSA BY PASSING BETWEEN TWO HEAD OF PRONATOR TERES.
- FOREARM – LIE BETWEEN FLEXOR DIGITORUM SUPERFICIALIS AND FLEXOR DIGITORUM PROFUNDUS.
- PALM – ENTER IN HAND LIE DEEP TO → FLEXOR RETINACULUM THROUGH CARPAL TUNNEL.

BRANCH –

1. MUSCULAR-ARM-PRONATOR TERES
   CUBITAL FOSSA → FLEXOR CARPI RADIALIS
   → FLEXOR DIGITORUM SUPERFICIALIS
   → PALMARIS LONGUS
   FOREARM → FLEXOR DIGITORUM PROFUNDUS
   → PRONATOR QUADRATUS
   → FLEXOR POLLICIS LONGUS
   PALM → THENAR MUSCLE
   → LATERAL 2 LUMBRICALS
2. CUTANEOUS - PALMARIS CUTANEOUS PALM
   (SUPPLY SKIN OVER THENAR MUSCLE)
3. ARTICULAR BRANCHES → ARE GIVEN TO ELBOW JOINT

CLINICAL ANATOMY

CARPAL TUNNEL SYNDROME

CAUSED BY COMPRESSION OF MEDIAL NERVE AND CARPAL TUNNEL

SENSORY CHANGE - LOSS OF SENSATION IN LATERAL 3 HALF DIGITS INCLUDING NAIL BEDS ON DORSUM OF HAND
- SKIN OVER THE THENAR EMINENCE IS NOT AFFECTED AS BRANCH OF MEDIAN NERVE SUPPLYING IT ARISE IN THE FOREARM

MOTOR CHANGE - APE / MONKEY LIKE THUMB DEFORMITY
- LOSS OF OPPOSITION OF THUMB, INDEX AND MIDDLE FINGER LAG BEHIND MAKING FIST DUE TO PARALYSIS OF FIRST AND SECOND LUMBRICAL MUSCLE.

VASOMOTOR CHANGE - SKIN AREA WITH SENSORY LOSS WARMER DUE TO ARTERIAL DILATATION.

TROPIC CHANGE - LONG STANDING CAUSE OF PARALYSIS LEAD TO DRYER SCALY SKIN.

FROMENTS SIGN (BOOK HOLDING TEST) – PATIENT IS UNABLE TO HOLD BOOK WITH THUMB AND OTHER FRINGERS

PAPER HOLDING TEST – PATIENT IS UNABLE TO HOLD PAPER BETWEEN THUMB AND FINGER

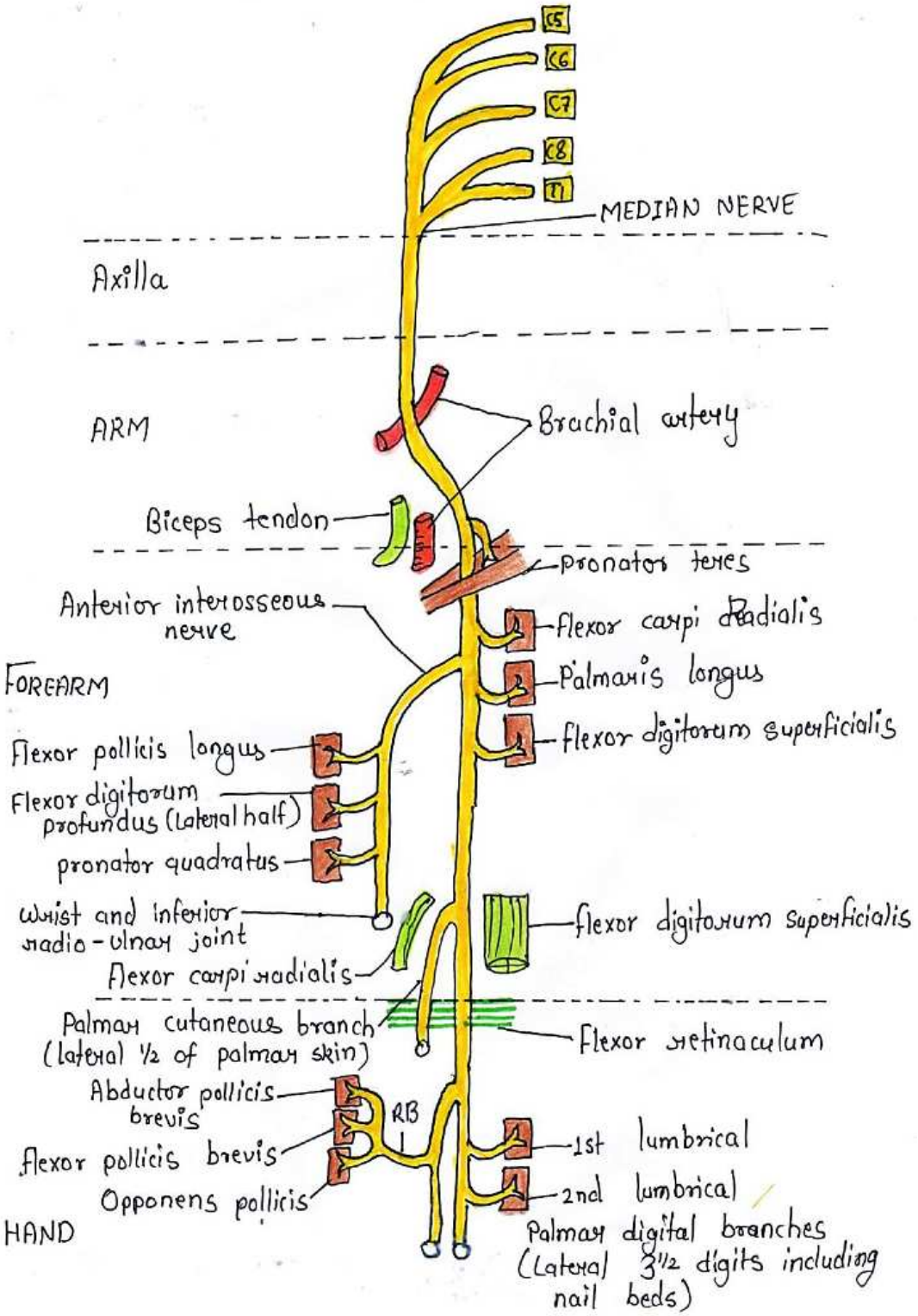
COMPLETE CLOW HAND

- **IF BOTH MEDIAN & ULNAR NERVE ARE PARALYSED**





CARPAL TUNNEL SYNDROME | COMPLETE CLOW HAND



ULNAR NERVE

ULNAR NERVE ALSO KNOWN AS MUSICIAN NERVE BECAUSE IT CONTROL FINE MOVEMENTS OF FINGERS.

ORIGIN - ARISE IN AXILLA FROM MEDIAL CORD OF BRACHIAL PLEXUS (C8 & T1)
- RECEIVE A CONTRIBUTION FROM VENTRAL RAMUS OF C7
- THE C7 FIBERS IN ULNAR NERVE SUPPLY FLEXOR CARPI ULNARIS

COURSE

- ARISE FROM BRACHIAL PLEXUS → DESCEND DOWN MEDIAL ASPECT OF UPPER ARM
- IN ARM ->MEDIAL SIDE TO BRACHIAL ARTERY UPTO MIDARM
  -> WHERE IT PIERCES → MEDIAL INTERMUSCULAR SEPTUM AND ENTER THE POSTERIOR COMPARTMENT OF ARM
- RUN DOWNAWARD TO BACK OF MEDIAL EPICONDYLE OF HUMEROUS WHERE IT EASILY PALPATED
- ENTER FOREARM → PASSING BETWEEN 2 HEAD OF FLEXOR CARPI ULNARIS
- IN FOREARM → LIE ALONG ---→ LATERAL BORDER OF FLEXOR CARPI ULNARIS.
  -> IN UPPER 1/3 FOREARM → RUN VERTICALLY DOWNWARD UNDER FLEXOR CARPI ULNARIS
  ->IN LOWER 2/3 FOREARM ->SUPERFICIAL AND LIE LATERAL TO FLEXOR CARPI ULNARIS
- ENTER PALM → PASSING SUPERFICIAL TO FLEXOR RETINACULUM THROUGH GUYON'S CANAL AND ENTER PALM

BRANCHES

1.MUSCULAR → FLEXOR CARPI ULNARIS
→ FLEXOR DIGITORUM PROFUNDUS (HALF)

2.CUTANEOUS → PALMER CUTANEOUS BRANCH
(SUPPLIES SKIN OVER HYPOTHENAR MUSCLE)
→ DORSAL CUTANEOUS BRANCH
(SUPPLIES MEDIAL 2 HALF FINGER DORSUM OF HAND)

3.PALMAR → SUPERFICIAL TERMINAL BRANCH → PALMARIS BREVIS
→ DIGITAL BRANCH
→ DEEP TERMINAL BRANCH - HYPOTHENAR
- INTEROSSEOUS MUSCLE
- MEDIAL 2 LUMBRICALS

CLINICAL ANATOMY

ULNAR CLAW HAND

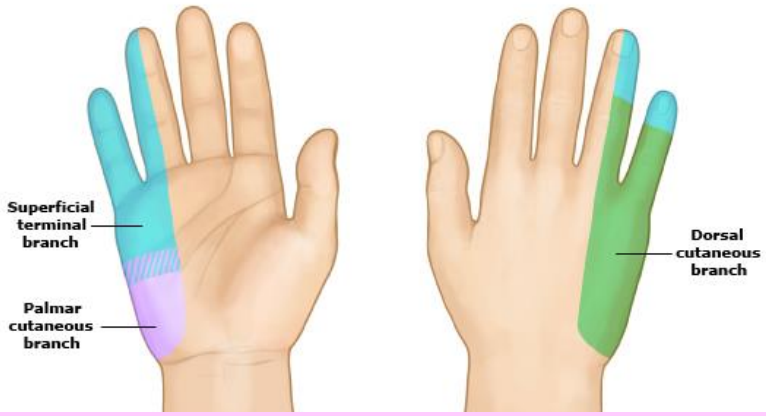
ULNAR NERVE INJURY AT WRIST -> PRODUCE ULNAR CLAW HAND

CAUSE - HYPEREXTENSION AT METACARPOPHALANGEAL JOINTS AND FLEXION AT THE INTERPHALANGEAL JOINTS, INVOLVE-INDEX OR MIDDLE FINGER.

SENSORY LOSS - MEDIAL 1/3RD OF PALM AND MEDIAL HALF OF FINGER.
- MEDIAL HALF OF DORSUM OF HAND.

VASOMOTOR CHANGE - SKIN AREA WITH SENSORY LOSS IS WARMER DUE TO ARTERIOLAR DILATATION

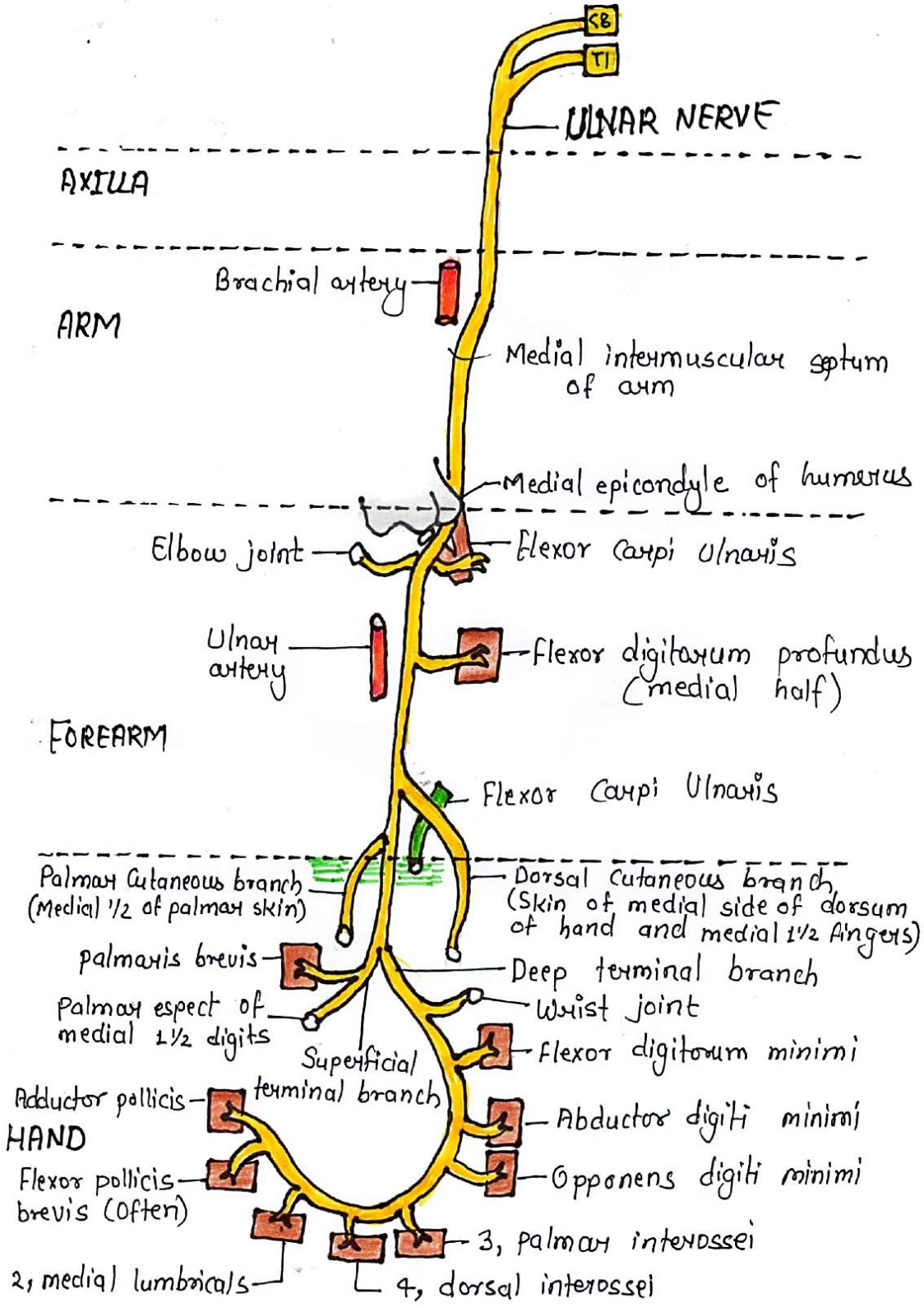
TROPIC CHANGE - LONG STANDING CASE OF PARALYSIS LEAD TO DRY AND SCALY SKIN



SENSORY LOSS IN ULNAR NERVE INJURY



ULNAR CLAW HAND



RADIAL NERVE

LARGEST NERVE OF UPPER LIMB AND LARGEST BRACNCH OF BRACHIAL PLEXUS

ORIGIN – RADIAL NERVE IS A CONTINUATION OF POSTERIOR CORD OF BRACHIAL PLEXUS IN AXILLA

ROOT VALUE – C5,6,7,8 & T1 (T1 FIBRES ARE NOT CONSTANT)

COURSE

AXILLA – RADIAL NERVE LIE POSTERIOR TO AXILLARY ARTERY
-THEN ENTER IN LOWER TRIANGULAR SPACE.

ARM – ENTER ARM → AT LOWER BORDER OF TERES MAJOR
- PASS BETWEEN LONG AND MEDIAL HEAD OF TRICEP TO ENTER LOWER TRIANGULAR SPACE

RADIAL SULCUS – RADIAL NERVE ENTER THROUGH LOWER TRIANGULAR SPACE INTO RADIAL SULCUS ALONG WITH PROFUNDA BRACHII ARTERY
- AT LOWER END OF RADIAL SULCUS → RADIAL NERVE PIERCES THE LATERAL INTERMUSCULAR SEPTUM OF ARM AND ENTERS ANTERIOR COMPARTMENT OF ARM
- HERE, IT FIRST DESCENDS BETWEEN BRACHIALIS AND BRACHIORADIALIS AND THEN BRACHIALIS & EXTERNSOR CARPI RADIALIS LONGUS THEN ENTER CUBITAL FOSSA

CUBITAL FOSSA – NERVE ENTER LATERAL SIDE OF CUBITAL FOSSA.
- TERMNATE INTO SUPERFICIAL & DEEP BRANCH.

DEEP BRANCH →SUPPLY →EXTENSOR CARPIRADIALIS BREVIS
→SUPINATOR.
THEN IT COURSE BETWEEN TWO HEAD OF SUPINATOR MUSCLE →REACH BACK OF FOREARM.

SUPERFICIAL BRANCH →LEAVE CUBITAL FOSSA TO ENTER LATERAL SIDE OF FRONT OF FOREARM.

DORSUM OF FOREARM – SUPERFICIAL BRANCH → DESCEND TILL SNUFF BOX
- TO REACH DORSUM OF HAND → SUPPLY SKIN OF DORSAL HALF HAND.

➢ BRANCHES

1.AXILLA → MUSCULAR → LONG HEAD OF TRICEP BRACHI
→ MEDIAL HEAD OF TRICEP BRACHI
→ CUTANEOUS → POSTERIOR CUTANEOUS NERVE OF ARM

2.RADIAL SULCUS → MUSCULAR → LATERAL HEAD OF TRICEP BRACHII
→ ANCONEAUS
→ CUTANEOUS → LATERAL CUTANEOUS NERVE OF ARM
→ POSTERIOR CUTANEOUS NERVE OF FOREARM.

3.LATERAL SIDE OF ARM—->MUSCULAR → BRACHIORADIAL
→ EXTENSOR CARPI RADIALIS LONGUS

4.DEEP BRANCH → DEEP BRANCH OF RADIAL NERVE
→ SUPINATOR MUSCLE
→ EXTENSOR CARPI RADIALIS BREVIS

5.SUPERFICIAL BRANCH → SUPERFICIAL BRANCH OF RADIAL NERVE
→ CUTANEOUS BRANCHES IN ANATOMICAL SNUFFBOX (TO SUPPLY LATERAL HALF OF DORSUM OF HAND)

CLINICAL ANATOMY

WRIST DROP

RADIAL NERVE INJURY IN RADIAL GROOVE DUE TO FRACTURE OF SHAFT OF HUMEROUS CAUSE WRIST DROP.

THIS RESULT IN WEAKNESS AND LOSS OF POWER OF EXTENSION IN WRIST.

WARTENBERG SYNDROME

COMPRESSION OF SUPERFICIAL BRANCH WHILE RUNNING BELOW THE BRACHIORADIALIS

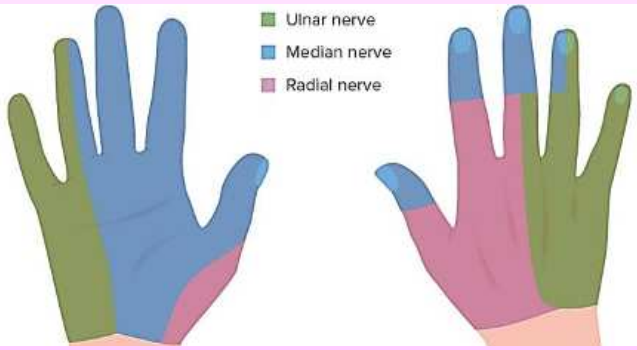
LOSS OF SENSATION IN LATERAL 2 AND ½ DIGIT IN DROSUM

SIGN – LITTLE FINGURE NOT ADDUCTED TOWARD RING FINGER & PALMAR INTEROSSEUS IS AFFECTED

RADIAL TUNNEL SYNDROME

COMPRESSION OF RADIAL NERVE AT ELBOW, NEUROPATHY OF DEEP RADIAL NERVE

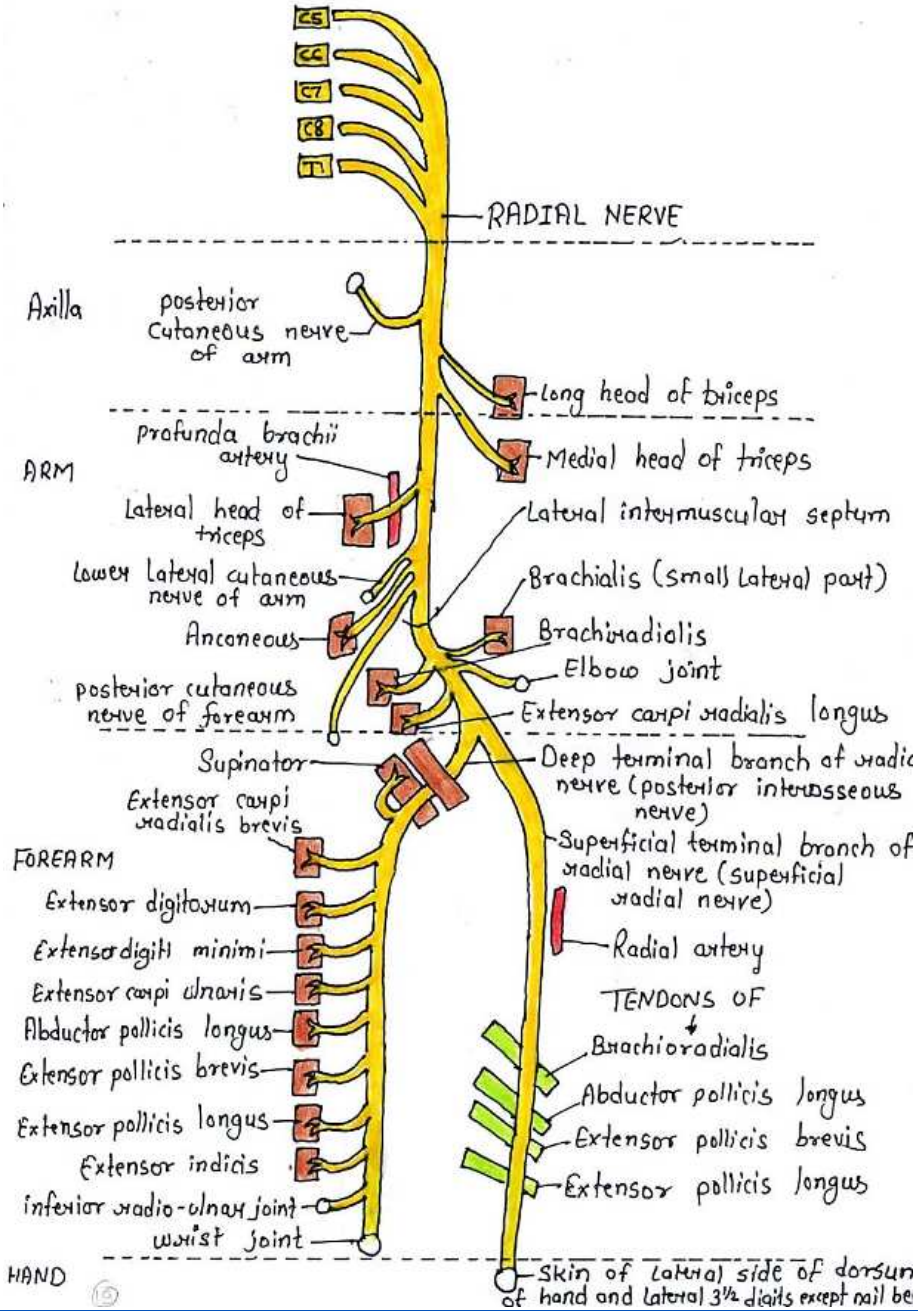
CHARACTERISTIC CLINICAL FEATURES – LOSS OF EXTENSION OF WRIST AND FINGERS BUT NO WRIST DROP



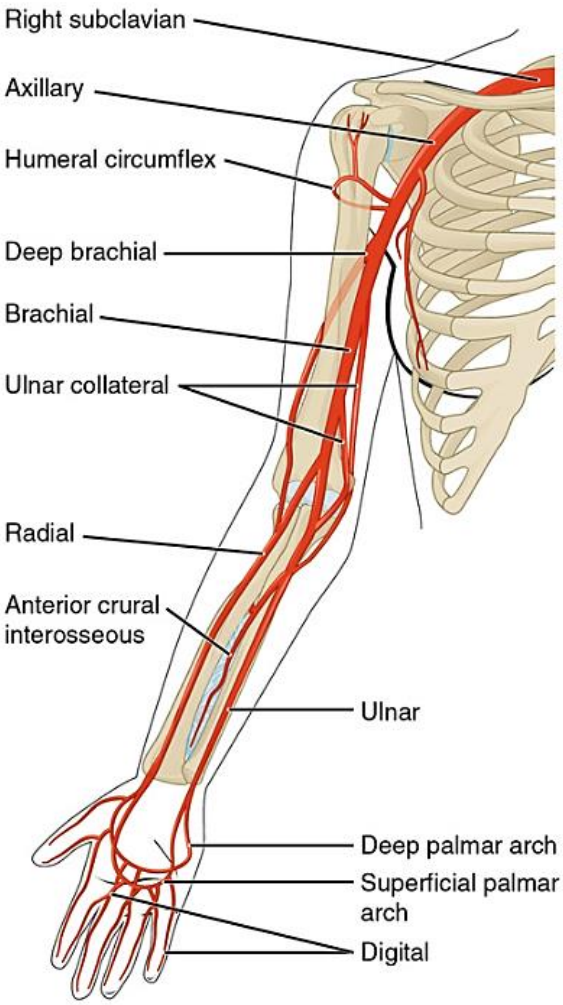
SENSORY INNERVATION OF HAND

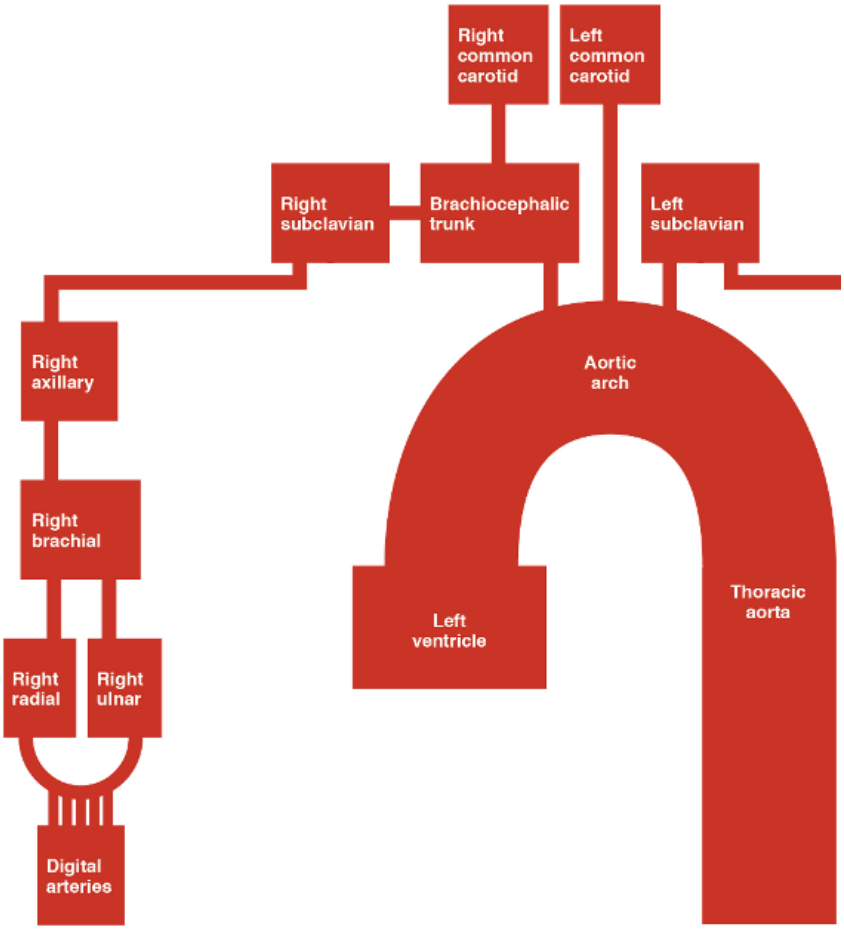


WRIST DROP

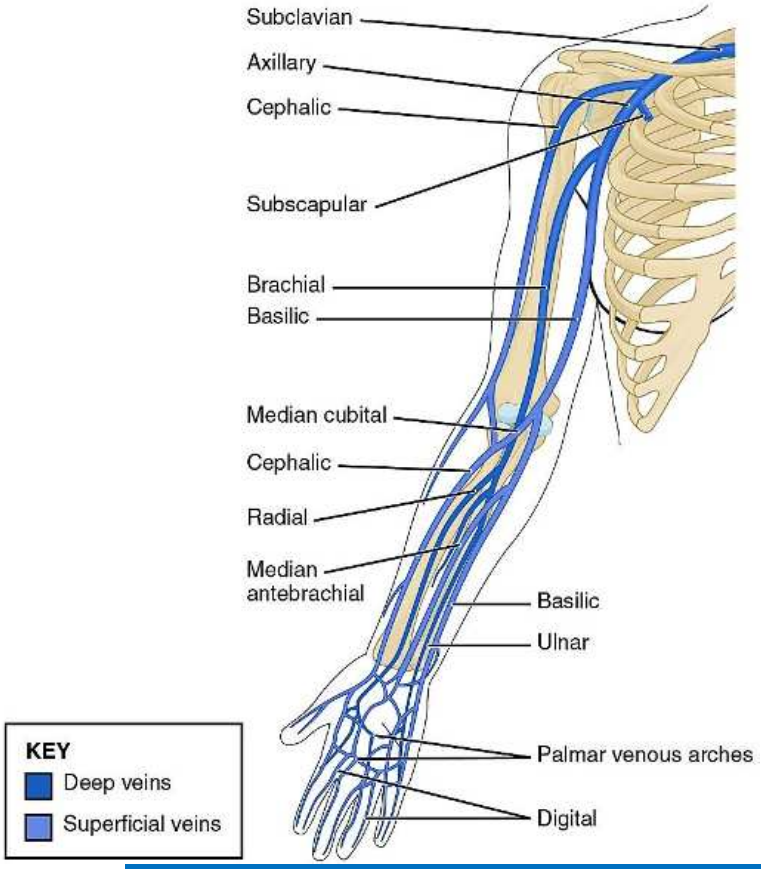


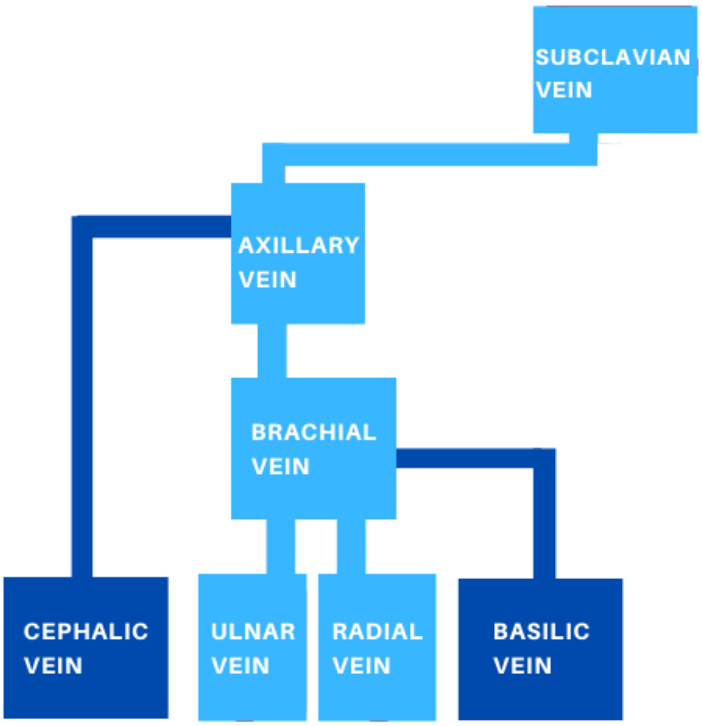
BLOOD SUPPLY OF UPPER LIMB





ARTERIAL SUPPLY OF UPPER LIMB





VENOUS DRAINAGE OF UPPER LIMB

AXILLARY ARTERY

INTRODUCTION – CONTINUATION OF SUB CLAVIAN ARTERY

-EXTEND OUTER BORDER OF FIRST RIB TO LOWER BORDER OF TERES MAJOR MS.

-DIVIDED —> 3 PARTS

- PECTORALIS MINOR MUSCLE CROSS ARTERY – DIVIDES INTO 3 PARTS

1ST PART – BRANCHES – SUPERIOR THORACIC ARTERY

2ND PART – BRANCHES – THORACOACROMIAL ARTERY

– LATERAL THORACIC ARTERY

3RD PART-BRANCHES – SUBSCAPULAR ARTERY

– ANTERIOR CIRCUMFLEX HUMERAL ARTERY

– POSTERIOR CIRCUMFLEX HUMERAL ARTERY

BRANCHES

1.SUPERIOR THORACIC ARTERY –

BRANCH – 1ST PART OF AXILIARY ARTERY

DIRECTION – DOWN, FORWARD & MEDIAL

SUPPLY – PECTORAL MAJOR

2.THORACOACROMIAL ARTERY –

BRANCH – 2ND PART OF AXILLARY ARTERY

3.LATERAL THORACIC ARTERY –

BRANCH – 2ND PART OF AXILLARY ARTERY

IT EMERGES – LOWER BORDER OF PECTORALIS MINOR

4.ANTERIOR CIRCUMFLEX HUMERAL ARTERY –

BRANCH –3RD PART OF AXILLARY ARTERY

ARISE – LOWER PART OF SUBSCAPULARIS

SUPPLY –ANASTOMOSIS POSTERIOR CIRCUMFLEX HUMERAL ARTERY

5.POSTERIOR CIRCUMFLEX HUMERAL ARTERY –

BRANCH –3RD PART OF AXILLARY ARTERY

EXTEND – LOWER BORDER OF SUBSCAPULARIS

SUPPLY –SHOULDER JOINT,DELTOID

6.SUBSCAPULARIS ARTERY –

BRANCH –3RD PART OF AXILLARY ARTERY

EXTEND –LOWER BORDER OF SUBSCAPULARIS TO TERMINATE NEAR INFERIOR ANGLE OF SCAPULA.

SUPPLY – LATISSIMUS DORSI & SERRATUS ANTERIOR MUSCLE.

RELATIONS

FIRST PART – ANTERIOR – PECTORALIS MAJOR & LOOP OF COMMUNICATON BETWEEN LATERAL & MEDIAL PECTORAL NERVE

– POSTERIOR – MEDIAL CORD OF BRCHIAL PLEXUS

– LONG THORACIC NERVE & SERRATUS ANTERIOR

– MEDAIL – AXILLARY VEIN

– LATERAL – LATERAL & POSTERIOR CORD OF BRACHIAL PLEXUS

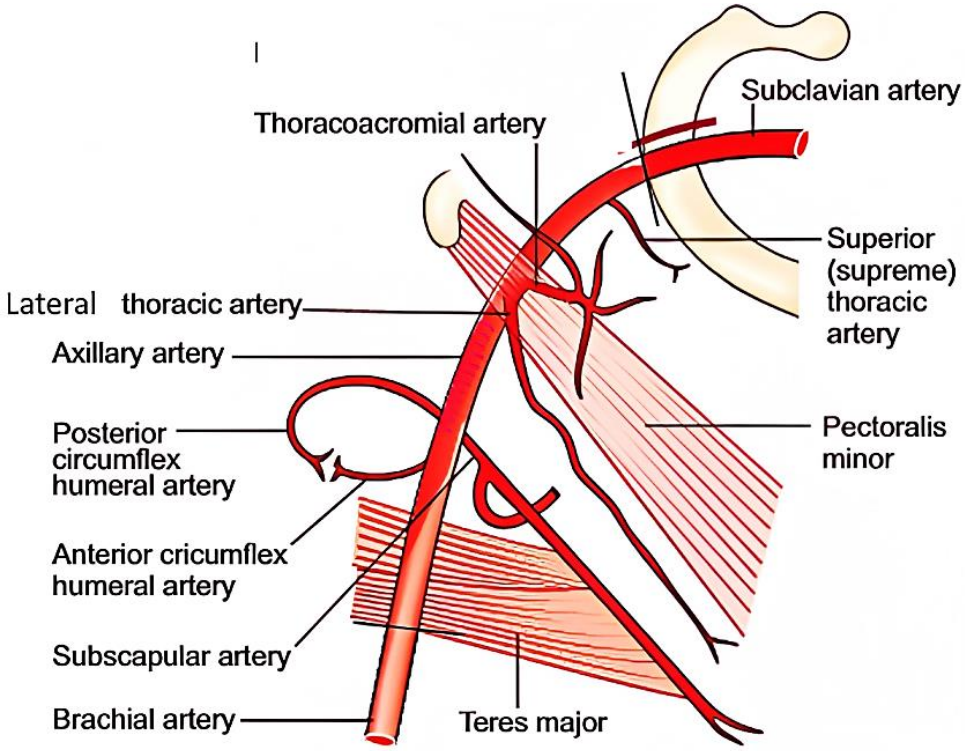
SECOND PART – ANTERIOR – PECTORALIS MINOR

– POSTERIOR – POSTERIOR CORD OF BRACHIAL PLEXUS

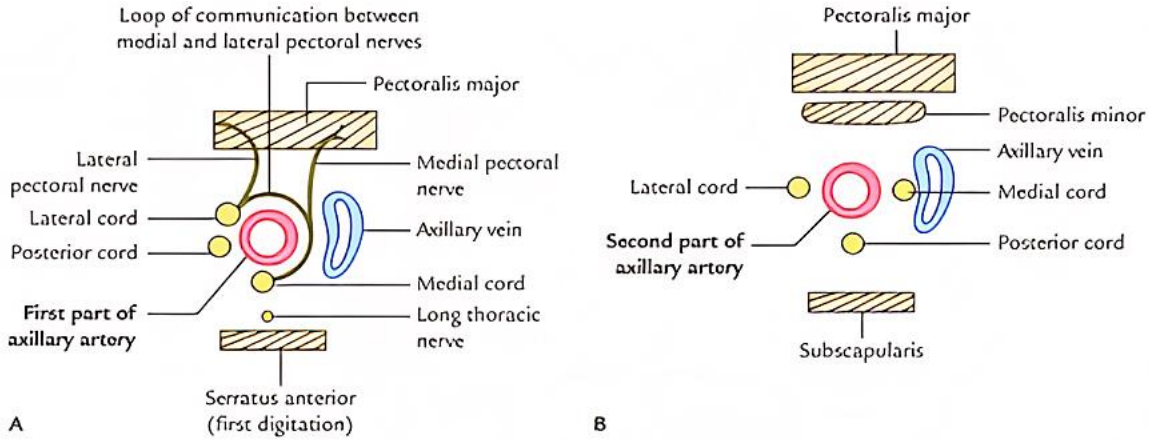
– SUBSCAPULARLIS

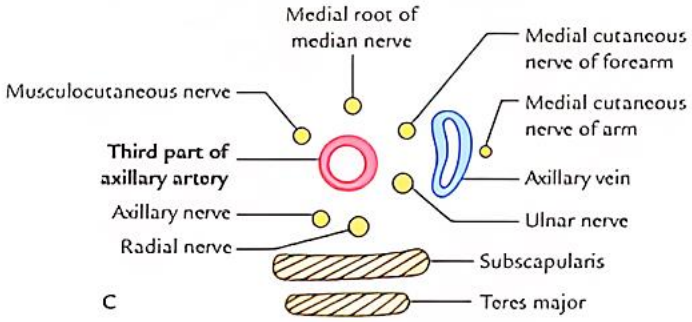
- MEDIAL – AXILLARY VEIN & MEDIAL CORD OF BRACHIAL PLEXUS
- LATERAL – LATERAL OF BRACHIAL PLEXUS

THIRD PART – ANTERIOR – MEDIAL ROOT OF MEDIAN NERVE
- POSTERIOR – RADIAL NERVE & AXILLARY NERVE
- SUBSCAPULARIS & TERES MAJOR
- MEDIAL – AXILLARY VEIN & ULNAR NERVE
- LATERAL – MUSCULOCUTANEOUS NERVE



BRACNHES OF AXILLARY ARTERY





BRACNHES OF AXILLARY ARTERY

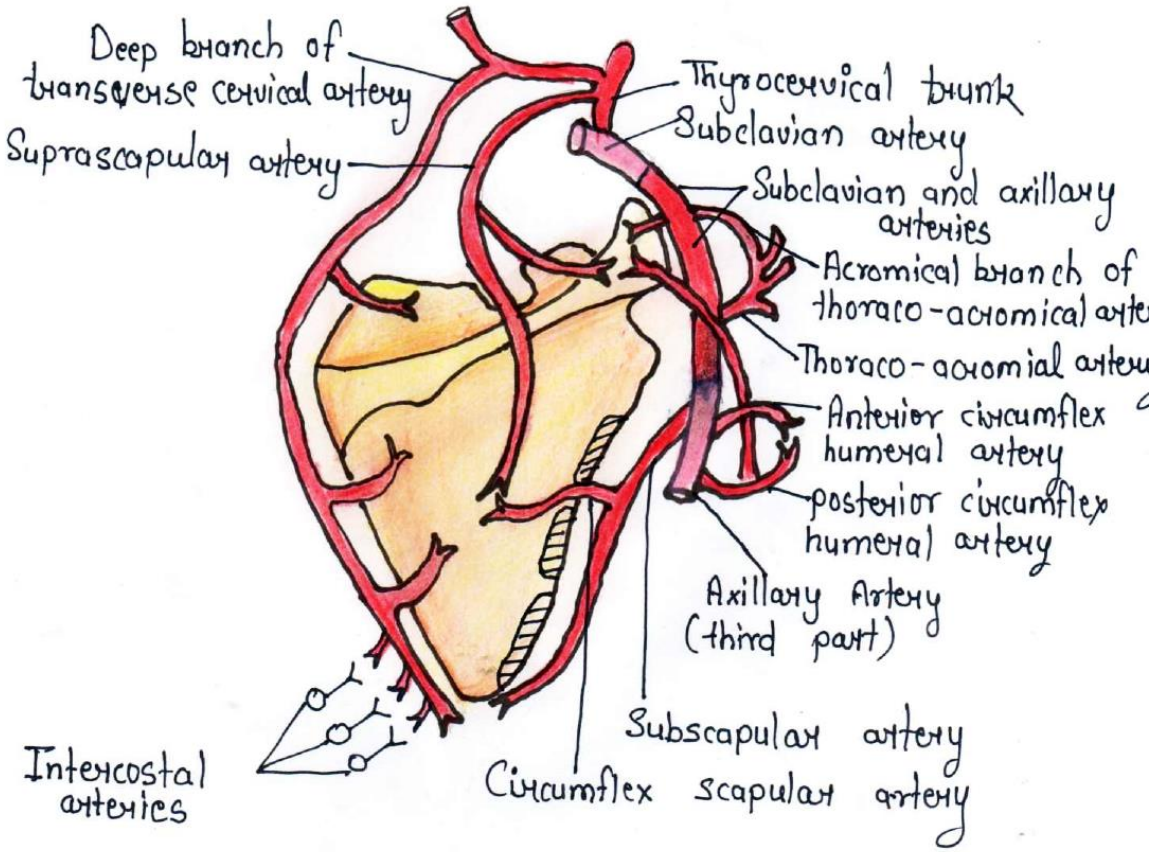
ANASTOMOSIS AROUND SCAPULA

ANSTOMOSIS AROUND SCAPULA IS BETWEEN BRANCH OF FRIST PART OF SUBCLAVIAN ARTERY & THIRD PART OF AXILLARY ARTERY

AREAS OF ANASTOMOSES

1) AROUND THE BODY OF SCAPULA –
   - (A) SUPRASCAPULAR ARTERY – BRACNH OF THYROCERVICAL TRUNK
   - (B) CIRCUMFLEX SCAPULAR ARTERY – BRANCH OF SUBSCAPULAR ARTERY
   - (C) DEEP BRANCH OF TRANSVERSE CERVICAL ARTERY – BRACNH OF THYROCERVICAL TRUNK

2) OVER THE ACROMION PROCESS –
   - (A) ACROMIAL BRANCH OF THORACO-ACROMIAL ARTERY
   - (B) ACROMIAL BRANCH OF SUPRASCAPULAR ARTERY
   - (C) ACROMIAL BRANCH OF POSTERIOR CIRCUMFLEX HUMERAL ARTERY



IMPORTANT INFORMATION

BRACHIAL ARTERY IS CONTINUATION OF AXILLARY ARTERY FROM LOWER BORDER OF TERES MAJOR MUSCLE
AT LEVEL OF NECK OF RADIUS IS DIVIDED INTO RADIAL AND ULNAR ARTERIES

BRANCHES OF BRACHIAL ARTERY

1) PROFUNDA BRACHII ARTERY -> GIVE RADIAL COLLATERAL ARTERY
2) SUPERIOR UNLAR COLLARERAL
3) INFERIOR ULNAR COLLARERAL

SUPERFICIAL & DEEP PALMAR ARCH

THE ARCH REPRESENTS AN IMPORTANT ANASTOMOSIS BETWEEN THE ULNAR AND RADIAL ARTERIES.

FORMATION

- THE SUPERFICIAL PALMAR ARCH IS FORMED AS THE DIRECT CONTINUATION OF THE ULNAR ARTERY BEYOND THE FLEXOR RETINACULUM, I.E., BY THE SUPERFICIAL PALMAR BRANCH.
- ON THE LATERAL SIDE: COMPLETED BY SUPERFICIAL PALMAR BRANCH OF RADIAL ARTERY.

RELATION

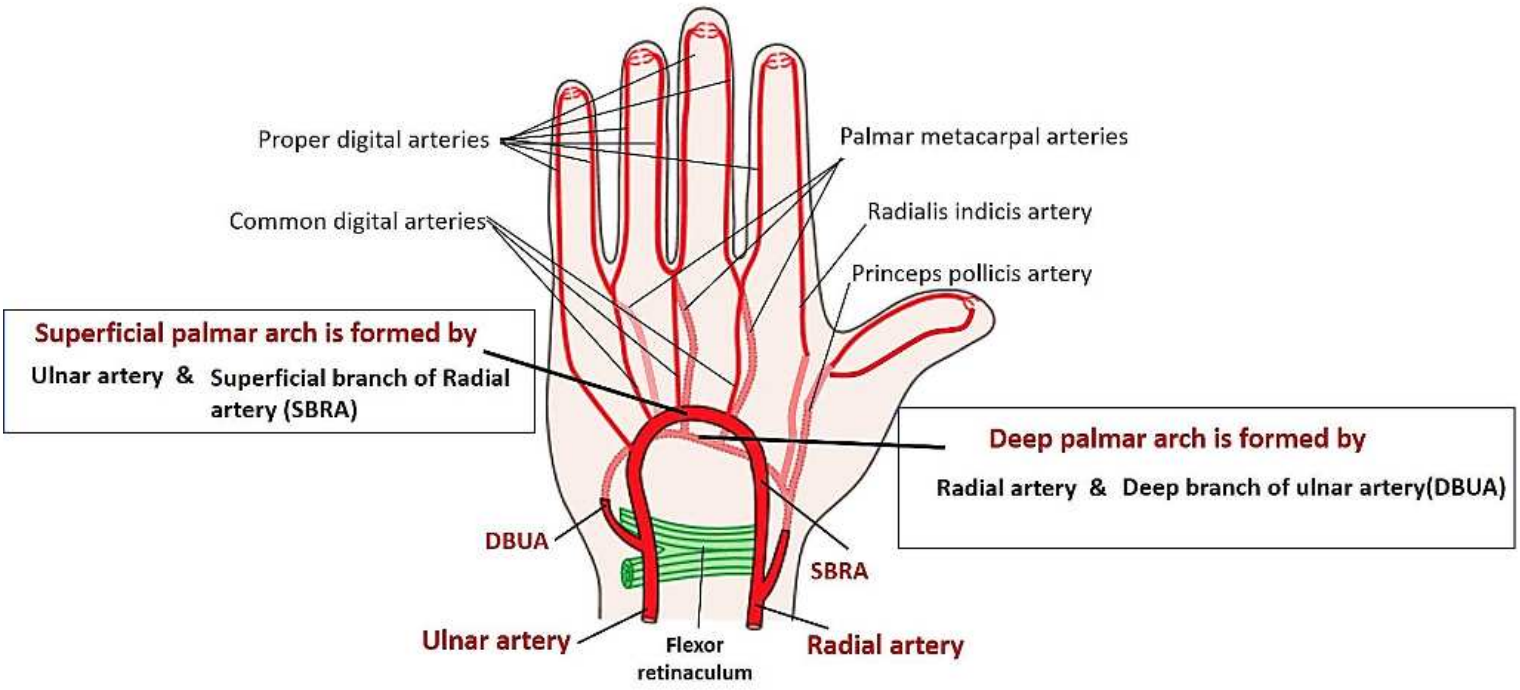
- THE SUPERFICIAL PALMAR ARCH LIES DEEP TO THE PALMARIS BREVIS AND THE PALMAR APONEUROSIS.
- IT CROSSES THE PALM OVER THE FLEXOR DIGITI MINIMI, THE FLEXOR TENDONS OF THE FINGERS, THE LUMBRICALS, AND THE DIGITAL BRANCHES OF THE MEDIAN NERVE.

BRANCHES

- SUPERFICIAL PALMAR ARCH GIVES OFF THREE COMMON DIGITAL AND ONE PROPER DIGITAL BRANCH WHICH SUPPLY THE MEDIAL 3½ DIGITS.

- THE LATERAL THREE COMMON DIGITAL BRANCHES ARE JOINED BY THE CORRESPONDING PALMAR METACARPAL ARTERIES FROM THE DEEP PALMAR ARCH.

- THE DEEP BRANCH OF THE ULNAR ARTERY ARISES IN FRONT OF THE FLEXOR RETINACULUM IMMEDIATELY BEYOND THE PISIFORM BONE. SOON IT PASSES BETWEEN THE FLEXOR AND ABDUCTOR DIGITI MINIMI TO JOIN AND COMPLETE THE DEEP PALMAR ARCH.



MAMMARY GLAND

BREAST IS MOST IMPORTANT STRUCTURE PRESENT IN PECTORAL REGION.
FOUND IN BOTH SEX BUT RUDIMENTARY IN MALE.
MODIFIED SWEAT GLAND.
SITUATION – LIE IN SUPERFICIAL FASCIA.
DIVIDE INTO 4 QUADRANT – UPPER MEDIAL, UPPER LATERAL, LOWER MEDIAL, LOWER LATERAL

- EXTENSION OF UPPER QUADRANT CALLED AXILLARY TAIL OF SPENCE ->WHICH PASSES THROUGH OPENING IN DEEP FASCIA.
- OPENING CALLED FORAMEN OF LANGER.

EXTENT OF BREAST –

VERTICALLY – 2ND – 6TH RIB
HORIZONTAL – LATERAL BORDER OF STERNUM TO MID AXILLARY LINE
DEEP RELATION – BREAST LIE ON DEEP FASCIA
-DEEPER THERE ARE PART OF 3 MUSCLES – PECTORALIS MAJOR
- SERRATUS ANTERIOR
- EXTERNAL OBLIQUE MUSCLE

STRUCTURE OF BREAST –

SKIN - CONICAL POSITION OF NIPPLE ->AT LEVEL->4TH INTERCOASTAL SPACE.
- NIPPLE PIERCED BY 15-20 LACTIFEROUS DUCT.
- IT CONTAIN CIRCULAR & LONGITUDINAL SMOOTH MUSCLE.
- SKIN SURROUNDING BASE OF NIPPLE IS PIGMENTED & FORM CIRCULAR AREA CALLED AREOLA

PARENCHYMA - MAMMARY GLAND COMPOUND TUBULOALVEOAR GLAND WHICH SECRETES MILK
- IT IS A MODIFIED SWEAT GLAND – GLAND CONSISTS 15-20 LOBES
- EACH LOBE -> CLUSTER OF ALVEOLI -> DRAINED BY LACTIFEROUS DUCT
- LACTIFEROUS DUCT -> CONVERGE -> TOWARD NIPPLE
- LATERAL DUCT NEAR ITS TERMINATION DILATATION -> CALLED LACTIFEROUS SINUS.

STROMA - FROM SUPPORTIVE FRAMEWORK OF GLAND
- PARTLY FIBROUS PARTLY FATTY

BLOOD SUPPLY

ARTERY ->
1.INTERNAL THORACIC ARTERY
2.LATERAL THORACIC, SUPERFICIAL THORACIC, ACROMIOTHORACIC
3.LATERAL BRANCH -> POSTERIOR INTERCOASTAL ARTERY

VEINS -> FOLLOW THE ARTERY
1.SUPERFICIAL VEIN -> DRAIN INTERNAL THORACIC VEIN
2.DEEP VEIN -> DRAIN-> AXILLARY & POSTERIOR INTERCOASTAL VEIN

NERVE SUPPLY

SUPPLY BY ANTERIOR AND LATERAL CUTANEOUS BRANCHES OF 4TH & 6TH INTERCOASTAL NERVE.

IMPORTANT INFORMATION

1) NERVE DO NOT CONTROL SECRETION OF MILK & SECRETION CONTROL BY HORMONE PROLACTIN
2) RETROMAMMARY SPACE – SPACE BETWEEN DEEP PART OF GLAND AND PECTORALIS FASCIA CONTAIN LOOSE CONNECTIVE TISSUE

LYMPHATIC DRAINAGE OF MAMMARY GLAND

LYMPHATIC DRAINAGE OF BREAST ASSUMED GREAT IMPORTANT TO SURGEON BECAUSE CARCINOMA SPREAD THROUGH LYMPH NODE.

LYMPH NODE -

1. AXILLARY LYMPH NODE – ANT., POST., LAT., CENTRAL, APCAL L.N. (75%)
2. INTERNAL MAMMARY L.N. – (20%)
3. SOME LYMPH (5%) -> FROM BREAST REACH-> * SUPRACLAVICULAR NODE
   * DELTOPECTORAL NODE
   * POST.INTERCOASTAL NODE
   * SUB DIAPHRAGMATIC

LYMPHATIC VESSELS

1) SUPERFICAL LYMPHATICS →DRAIN→SKIN OVER BREAST EXCEPT NIPPLE & AREOLA
2) DEEP LYMPHATIC --→DRAIN-→PARENCHYMA OF BREAST
3) LYMPH FROM ANTERIOR & POSTERIOR LYMPH NODE PASS → CENTRAL & LATERAL LYMPH NODE →THROUGH →APICAL GROUP →SUPRACLAVICAL NODES
4) INTERNAL MAMMARY NODE DRAIN LYMPH NOT ONLY FROM INNER HALF OF BREAST BUT ALSO OUTER HALF OF BREAST.
5) PLEXUS OF LYMPHATIC VESSELS PRESENT DEEP TO AREOLA IS SUBAEROLAR PLEXUS OF SAPPEY →DRAIN → ANTERIOR LYMPH NODE.
6) DEEP LYMPHATIC OF BREAST → PASS THROUGH PECTORALIS MAJOR MUSCLE & CLAVIPECTORAL FASICA→ APICAL NODE.

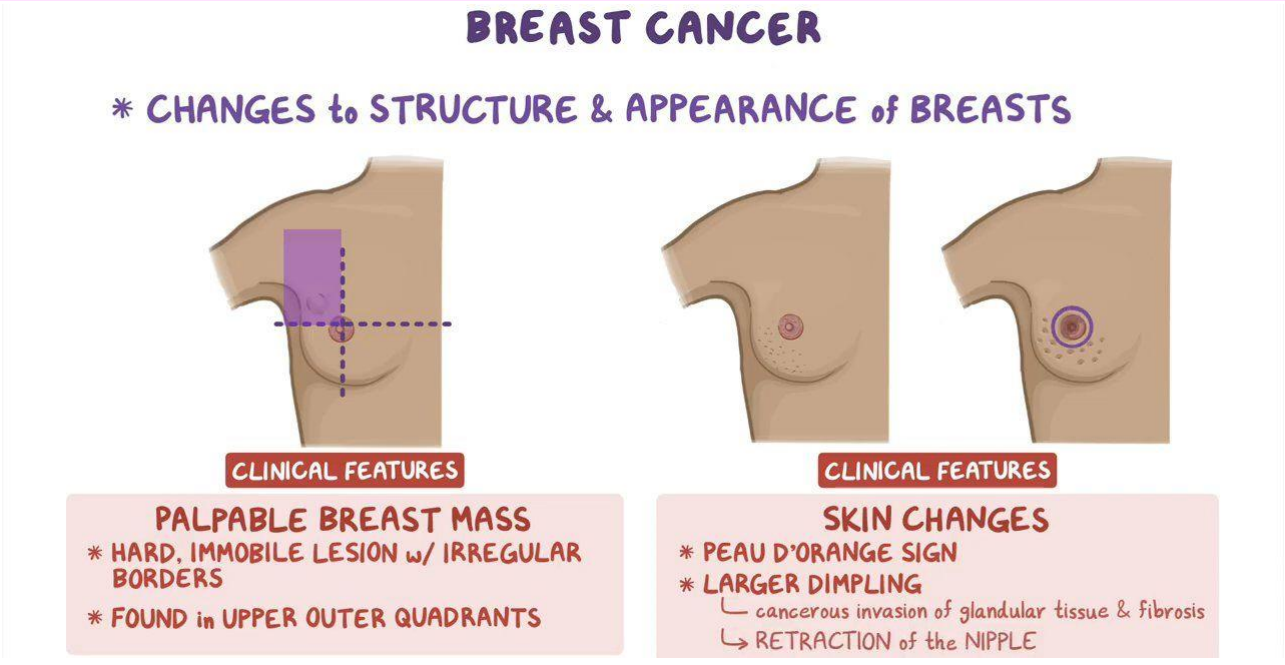
CLINICAL ANATOMY

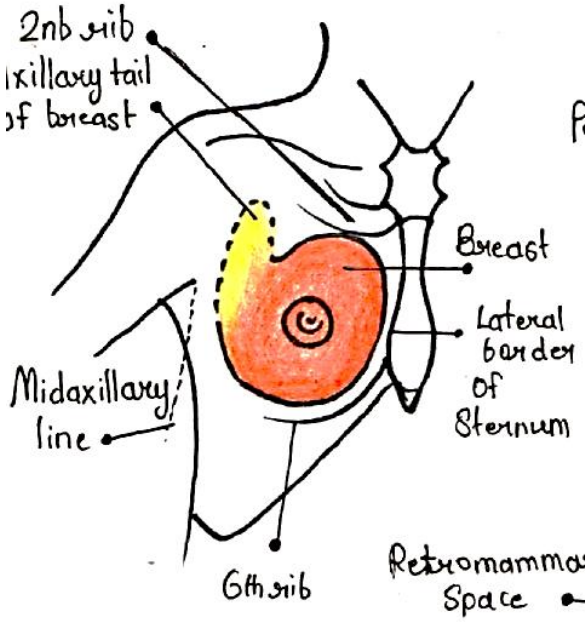
1) UPPER AND LOWER QUADRANT OF BREAST IS FREQUENT SITE OF CARCINOMA

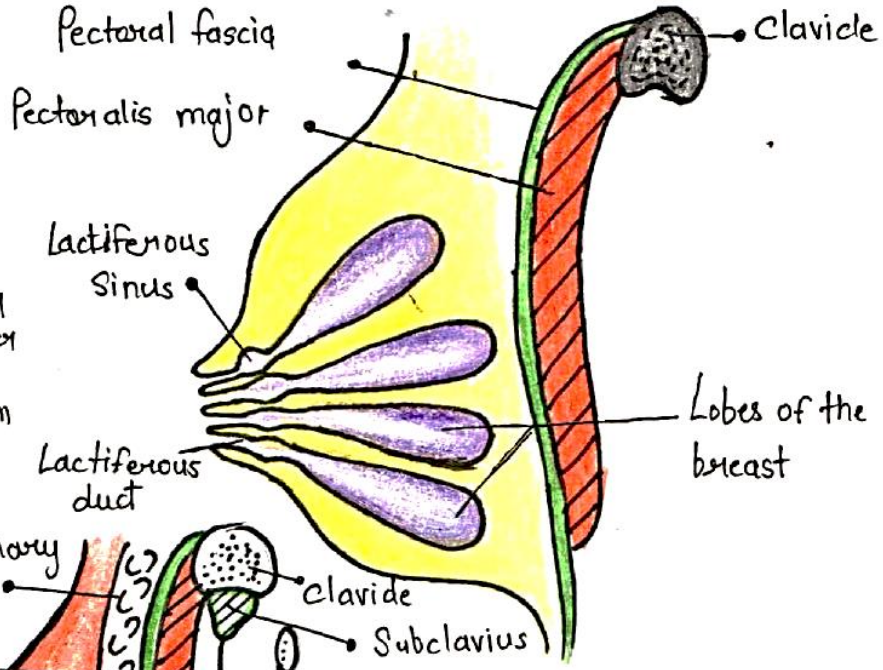
2) CANCER CELL MAY INFILTRATE THE SUSPENSORY LIGAMENT THEN BREAST BECOMES FIX CONTRACTION OF LIGAMENT →CAUSE RETRACTION OR FOLDING OF SKIN.

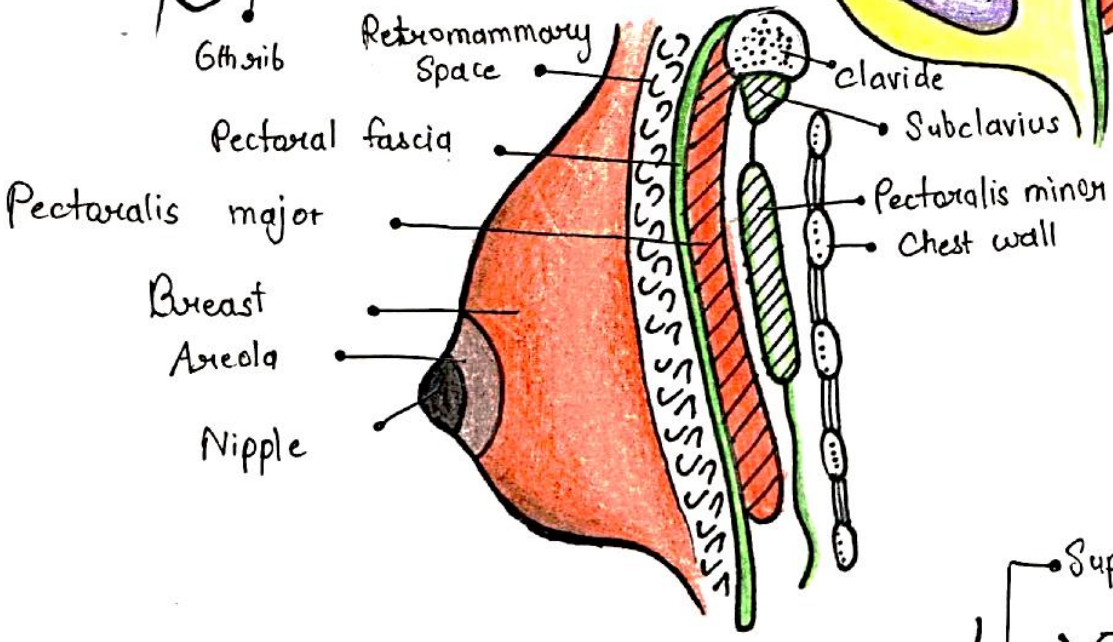
3) OBSTRUCTION OF SUPERFICIAL LYMPH VESSEL BY CANCER CELL →PRODUCE OEDEMA OF SKIN →APPEAR GIVING APPEARANCE LIKE SKIN OF ORANGE *(PEAU D'ORANGE APPEARANCE).*

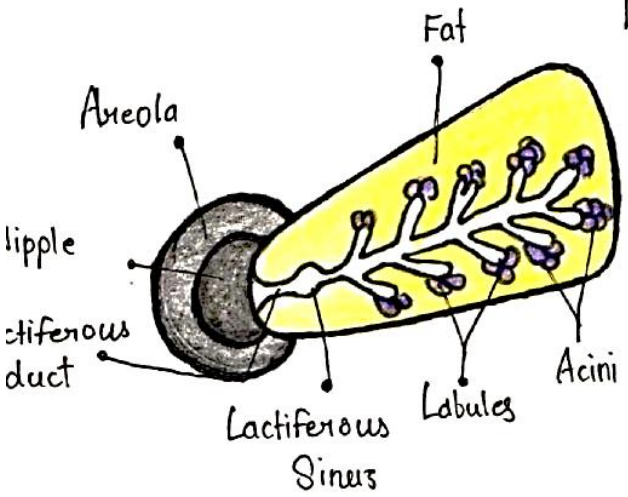
4) CARCINOMA USUALLY ARISES FROM EPITHELIUM OF LARGE DUCTS.

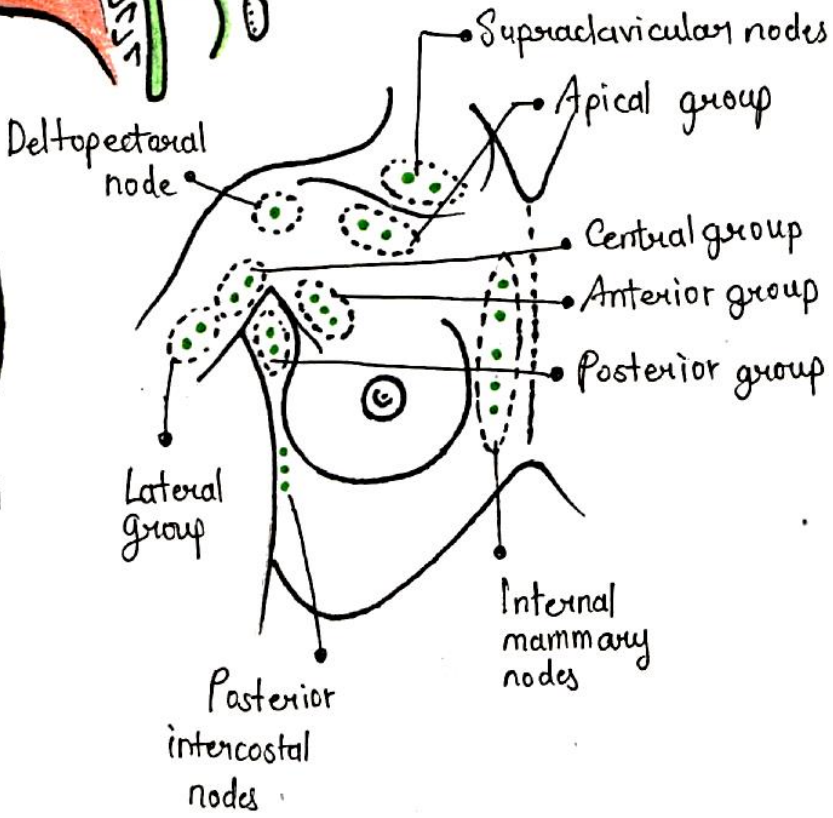



2nd rib
Axillary tail of breast
Breast
Lateral border of Sternum
Midaxillary line
6th rib


Pectoral fascia
Clavicle
Pectoralis major
Lactiferous Sinus
Lobes of the breast
Lactiferous duct

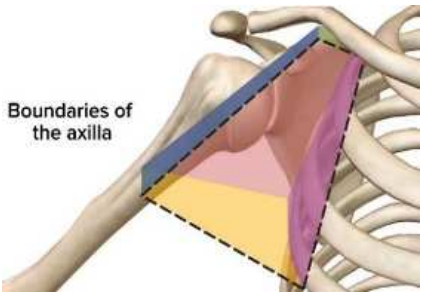

Retromammary Space
Clavide
Subclavius
Pectoral fascia
Pectoralis minor
Pectoralis major
Chest wall
Breast
Areola
Nipple


Fat
Areola
lipple
ctiferous duct
Lactiferous Sinus
Labules
Acini


Supraclavicular nodes
Apical group
Deltopectoral node
Central group
Anterior group
Posterior group
Lateral Group
Internal mammary nodes
Posterior intercostal nodes

AXILLA

DEFINITION - PYRAMIDAL SPACE,
- SITUATED – UPPER PART OF THE ARM AND CHEST WALL

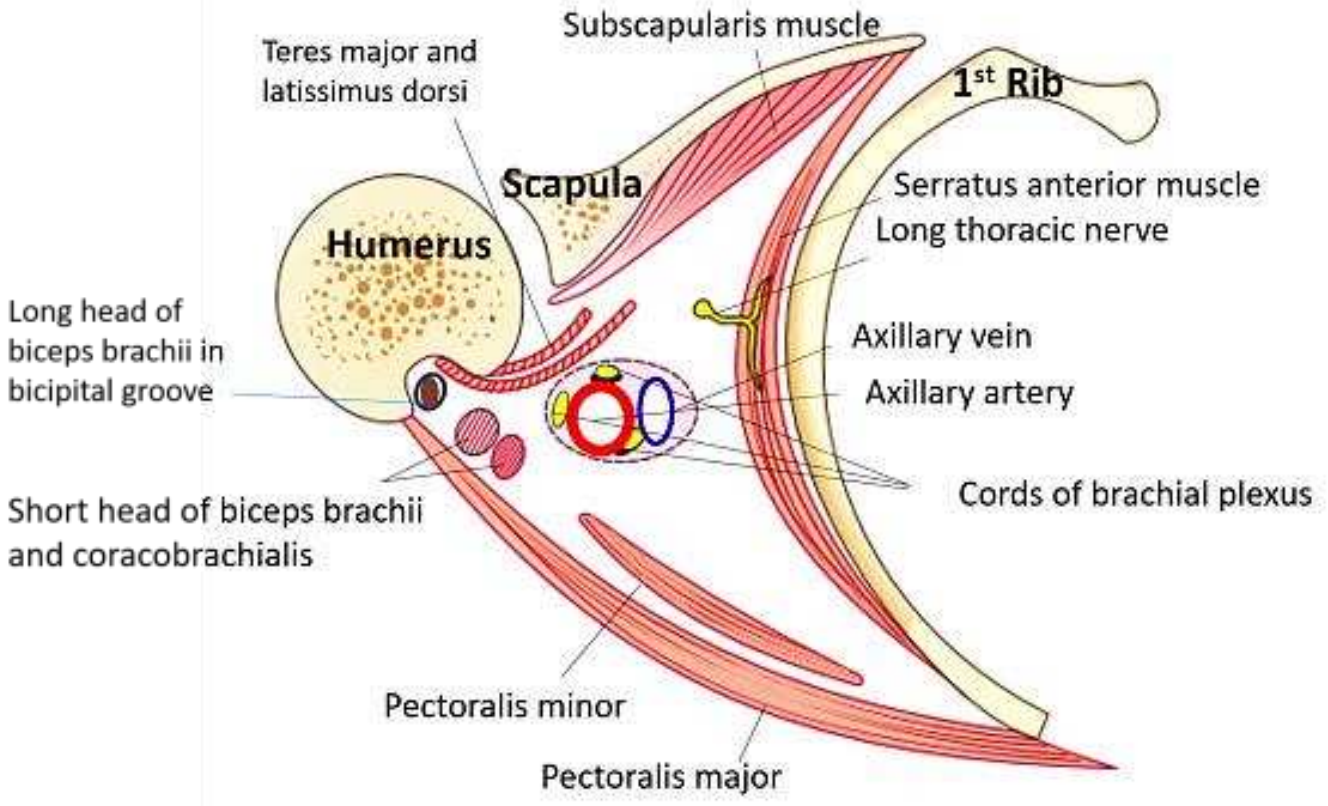


BOUNDARIES –
APEX – ANTERIOR COVER BY CLAVICLE
- POSTERIOR BY SUPERIOR BORDER OF SCAPULA
- MEDIAL BY OUTER BORDER OF FIRST RIB
BASE – SKIN, SUPERFICIAL FASCIA
ANTERIOR WALL – PECTORAL MAJOR & MINOR, CLAVIPECTORAL FASCIA
POSTERIOR WALL – SUBSCAPULARIS, TERES MAJOR , LATISSIMUS DORSI
LATERAL WALL – NARROW, CORACOBRACHIALIS & SHORT HEAD OF BICEP BRACHII

CONTENTS -
1) AXILLARY ARTERY & VEIN BRANCHES
2) INFRACLAVICULAR PART OF BRACHIAL PLEXUS
3) AXILLARY LYMPH NODE
4) LONG THORACIC & INTERCOSTOBRACHIAL NERVE
5) AXILLARY FAT AND AREOLAR TISSUE

IMPORTANT INFORMATION

CLAVIPECTORAL FASCIA
- IS A FIBROUS SHEET SITUATED DEEP TO CLAVICULAR PORTION OF PECTORALIS MAJOR MUSCLE
- ITS EXTEND FROM CLAVICLE ABOVE TO AXIALLARY FASCIA BELOW
- PIERCED BY - LATERAL PECTORAL NERVE
- CEPHALIC VEIN
- THORACOACROMIAL ARTERY



CUBITAL FOSSA

INTRODUCTION - TRIANGULAR HOLLOW, SITUATED - FRONT OF ELBOW
- BASE -> HORIZONTAL & DIRECTED UPWARD
- APEX -> DIRECTED DOWNWARD

"CORRESPONDS TO LOWER HALF OF POPLITEAL FOSSA OF LOWRER LIMB"

BOUNDARIES

1. LATERAL - MEDIAL BORDER OF BRACHIORADIALIS
2. MEDIAL - LATERAL BORDER OF PRONATOR TERES
3. BASE - IMAGINARY LINE, JOINING TWO EPICONDYLES OF HUMERUS
4. APEX - AREA WHERE - BRACHIORADIALIS & PRONATOR TERES CROSSES
5. ROOF - SKIN, SUPERFICIAL FASCIA
   - BICIPITAL APONEUROSIS
6. FLOOR - BRACHIALIS

CONTENTS

1) MEDIAN NERVE - GIVE BRANCHES -FLEXOR CARPI RADIALIS
   -PALMARIS LONGUS, FLEXOR DIGITORUM SUPERFICIALIS
2) RADIAL NERVE - GIVE - POSTERIOR INTEROSSEI NERVE AND DEEP BRANCH OF RADIAL NERVE
3) BRACHIAL ARTERY - GIVE - RADIAL ARTERY -> RADIAL RECURRENT BRANCH
   - ULNAR ARTERY -> ANTERIOR & POSTERIOR RECURRENT BRANCH & COMMON INTEROSSEOUS BRANCH
4) TENDON OF BICEP BRACHII

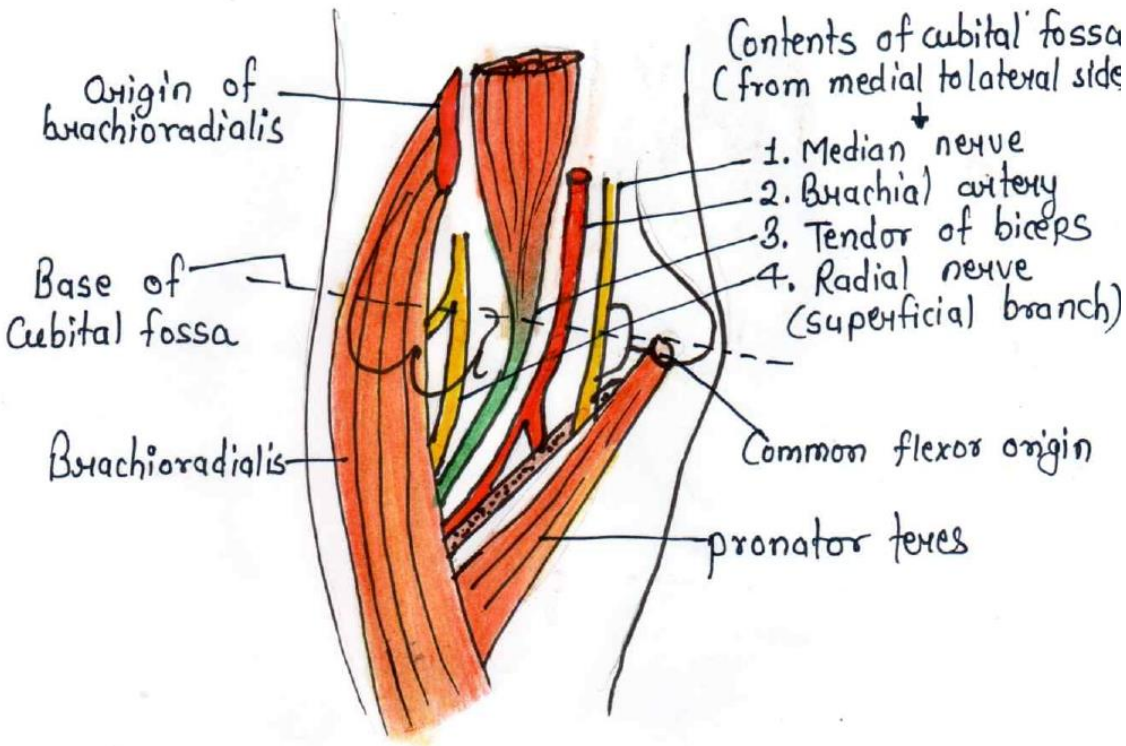
CLINICAL ANATOMY

MEDIAN CUBITAL VEIN - VEIN OF CHOICE FOR INTRAVENOUS INJECTION.

BRACHIAL ARTERY -BLOOD PRESSURE UNIVERSALLY RECORDED - BRACHIAL ARTERY INFRONT OF ELBOW

MNEMONIC

MBBS (M-MEDIAN NERVE, B - BRACHIAL ARTERY, B - BICEPS TENDON, S - SUPERFICIAL RADIAL NERVE



DELTOID MUSCLE

ORIGIN WITH FEATURES

1. THE ANTERIOR BORDER AND ADJOINING SURFACE OF THE LATERAL ONE-THIRD OF THE CLAVICLE.
2. THE LATERAL BORDER OF THE ACROMION PROCESS WHERE FOUR SEPTA OF ORIGIN ARE ATTACHED.
3. LOWER LIP OF THE CREST OF THE SPINE OF THE SCAPULA.

THE ACROMIAL PART OF DELTOID IS AN EXAMPLE OF A MULTIPENNATE MUSCLE. MANY FIBRES ARISE FROM FOUR SEPTA OF ORIGIN THAT ARE ATTACHED ABOVE TO THE ACROMION PROCESS.

INSERTION

THE V-SHAPED DELTOID TUBEROSITY OF THE HUMERUS WHERE THREE SEPTA OF INSERTION ARE ATTACHED

NERVE SUPPLY -> AXILLARY NERVE (C5, C6).

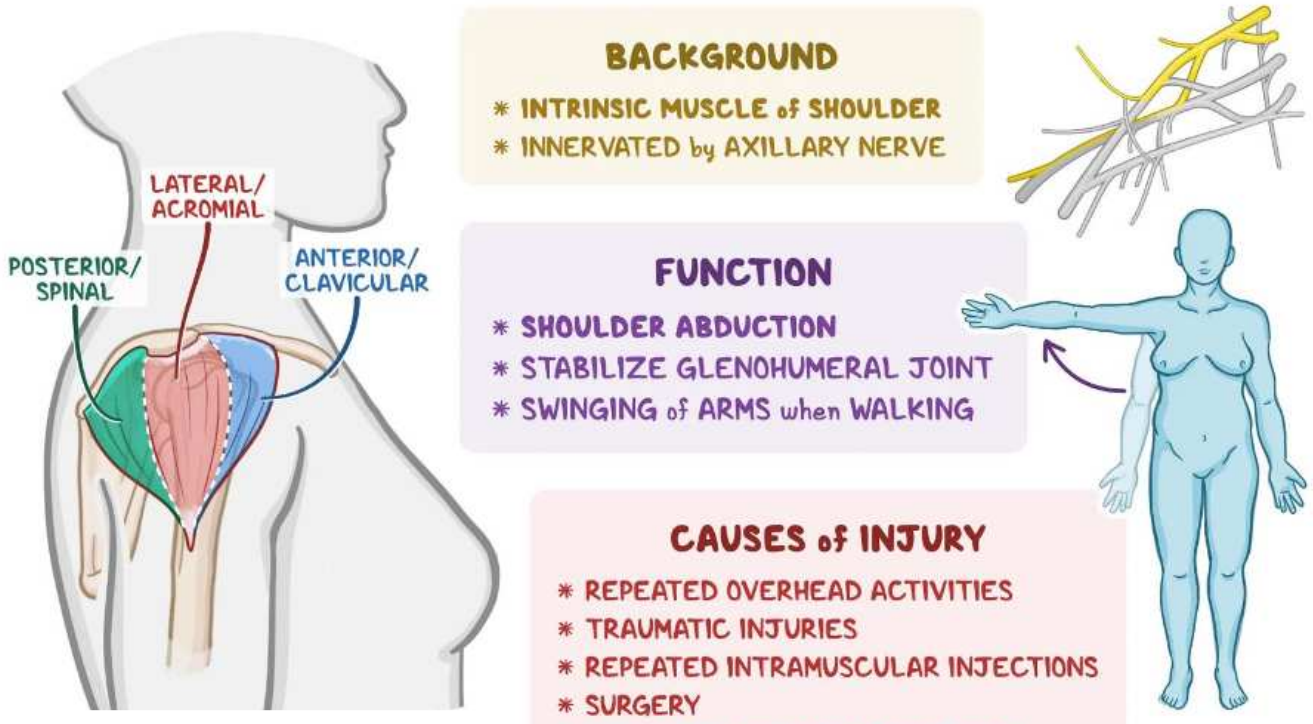
ACTIONS

1. THE MULTIPENNATE ACROMIAL FIBRES ARE POWERFUL ABDUCTORS OF THE ARM AT THE SHOULDER JOINT FROM BEGINNING TO 90°.
2. THE ANTERIOR FIBRES ARE FLEXORS AND MEDIAL ROTATORS OF THE ARM.
3. THE POSTERIOR FIBRES ARE EXTENSORS AND LATERAL ROTATORS OF THE ARM
4. MIDDLE FIBRES ARE STRONG ABDUCTOR OF ARM FROM 15° TO 90°

CLINICAL ANATOMY

SITE OF INTRAMUSCULAR INJECTION IN DELTOID

INTRAMUSCULAR INJECTION ARE COMMONLY GIVEN IN LOWER HALF OF THE DELTOID TO AVOID INJURY TO AXILLARY NERVE, WHICH WINDS AROUND SURGICAL NECK OF HUMERUS



ROTATOR CUFF

1.FIBROUS SHEATH→FORM BY 4 FLATTENED TENDONS WHICH BLEND WITH THE CAPSULE OF SHOULDER JOINT AND STRENGTHENED IT.

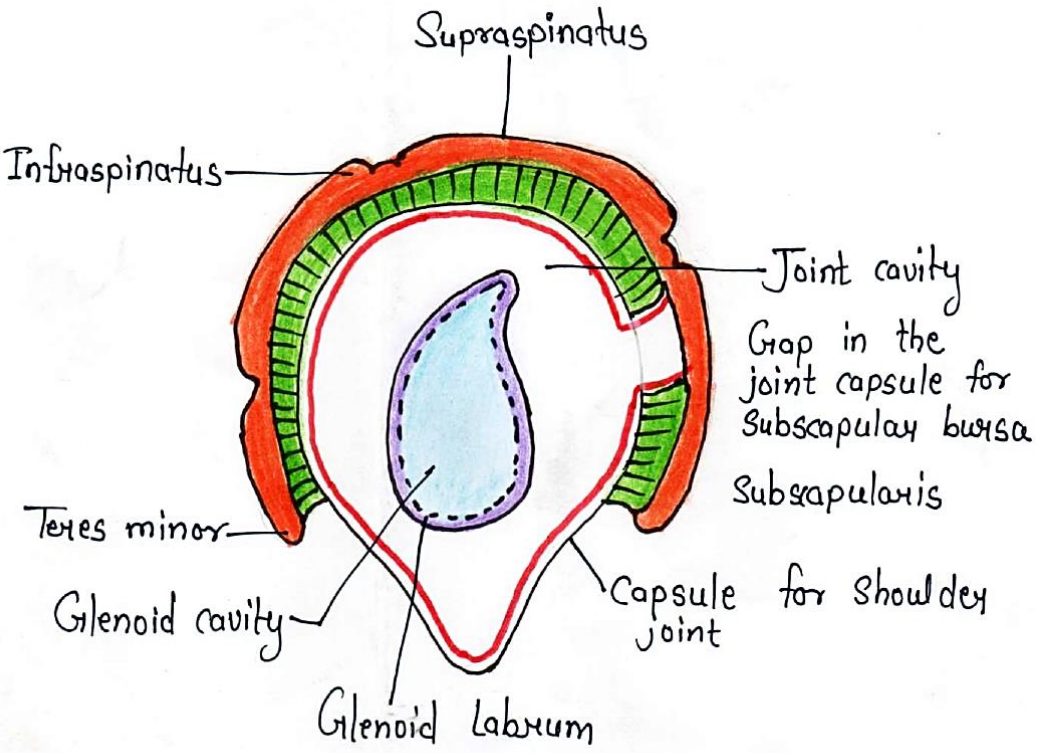
2.MUSCLE WHICH FORM ROTATOR CUFF ARISE → SCAPULA
INSERTION→ LESSER & GREATER TUBERCLE HUMEROUS

3.MUSCLE FORM ROTATOR CUFF → SUBSCAPULARIS, SUPRASPINATUS, INFRASPINATUS, TERES MINOR.

4.CUFF GIVE STRENGTH TO SHOULDER JOINT → EXCEPT→ INFERIORLY.

5.THUS DISLOCATION OF HUMEROUS OCCUR COMMONLY INFERIOLATERAL.

6.SUBACROMIAL BURSA → FLUID FILLED SAC. REDUCED FRICTION
→ LARGEST BUSRA OF BODY - BLOW CORACOACROMIAL ARC.



INTERMUSCULAR SPACES IN SCAPULAR REGION

INTERMUSCLUAR SPACES KNOWLEDGE IS ESSENTIAL DURING SURGERY IN SHOULDER REGIONPOTENTIAL SPACES ARE FORMED BETWEEN:

1) LATERAL BORDER SCAPULA
2) MEDIAL ASPECT OF HUMERUS
3) LONG HEAD OF TRICEPS BRACHII
4) TERES MINOR OR SUBSCAPULARIS
5) TERES MAJOR MUSCLES.

- IN THE UPPER PART - QUADRANGULAR SPACE LATERALLY
  - UPPER TRIANGULAR SPACE MEDIALLY.
- IN THE LOWER PART - LOWER TRIANGULAR SPACE.

QUADRANGULAR SPACE

BOUNDARIES – SUPERIOR - SUBSCAPULARIS IN FRONT.
- CAPSULE OF THE SHOULDER JOINT.
- INFERIOR BORDER OF TERES MINOR BEHIND.

- INFERIOR - SUPERIOR BORDER OF TERES MAJOR.
- MEDIAL - LATERAL BORDER OF LONG HEAD OF THE TRICEPS BRACHII.
- LATERAL - SURGICAL NECK OF THE HUMERUS

CONTENTS - AXILLARY NERVE
- POSTERIOR CIRCUMFLEX HUMERAL VESSELS

UPPER TRIANGULAR SPACE

BOUNDARIES – SUPERIOR-> INFERIOR BORDER OF TERES MINOR.
- LATERAL -> MEDIAL BORDER OF LONG HEAD OF THE TRICEPS BRACHII.
- INFERIOR -> SUPERIOR BORDER OF TERES MAJOR.

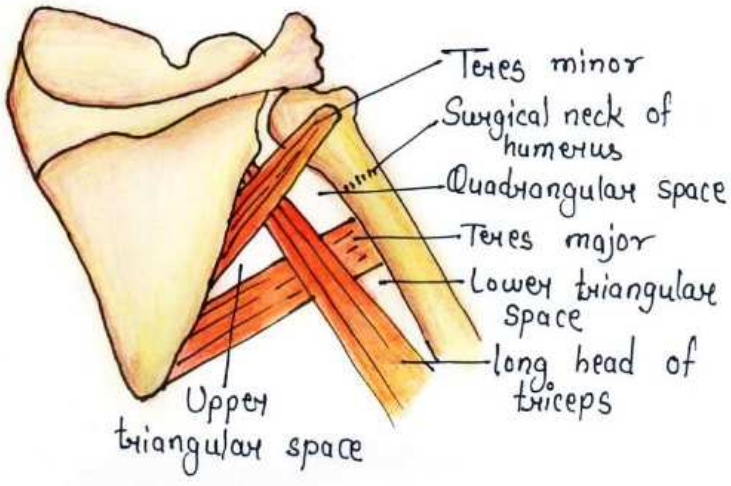
CONTENTS – CIRCUMFLEX SCAPULAR ARTERY.

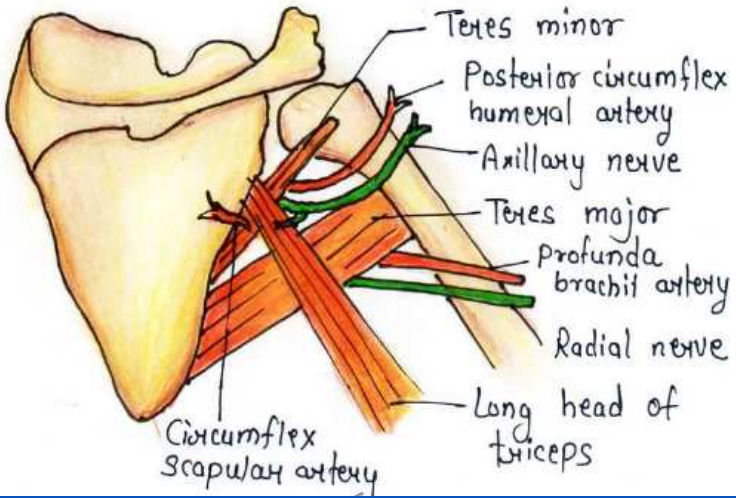
LOWER TRIANGULAR SPACE

IT IS DIAGONALLY OPPOSITE THE UPPER TRIANGULAR SPACE.

BOUNDARIES – MEDIAL -> LATERAL BORDER OF LONG HEAD OF THE TRICEPS BRACHII.
- LATERAL -> MEDIAL BORDER OF HUMERUS.
- SUPERIOR -> LOWER BORDER OF TERES MAJOR

CONTENTS - RADIAL NERVE
- PROFUNDA BRACHII VESSELS





FLEXOR RETINACULUM

STRONGEST FIBROUS BAND WHICH BRIDGE ANTERIOR CONCAVITY OF CARPUS & CONVERT INTO TUNNEL → CARPAL TUNNEL.

ATTACHMENT – MEDIAL → PISIFORM BONE, HOOK OF HAMATE
LATERAL → TUBERCLE OF SCAPHOID, CREST OF TRAPEZIUM.

MEDIAL SUPERFICIAL SLIP → PISIFORM BONE
(CONVERT INTO TUNNEL-→PASS-→ULNAR VESSEL & NERVE)

LATERAL DEEP SLIP → MEDIAL GROOVE ON TRAPEZIUM
(CONVERT INTO TUNNEL--→PASS--→FLEXOR CARPI RADIALIS TENDON)

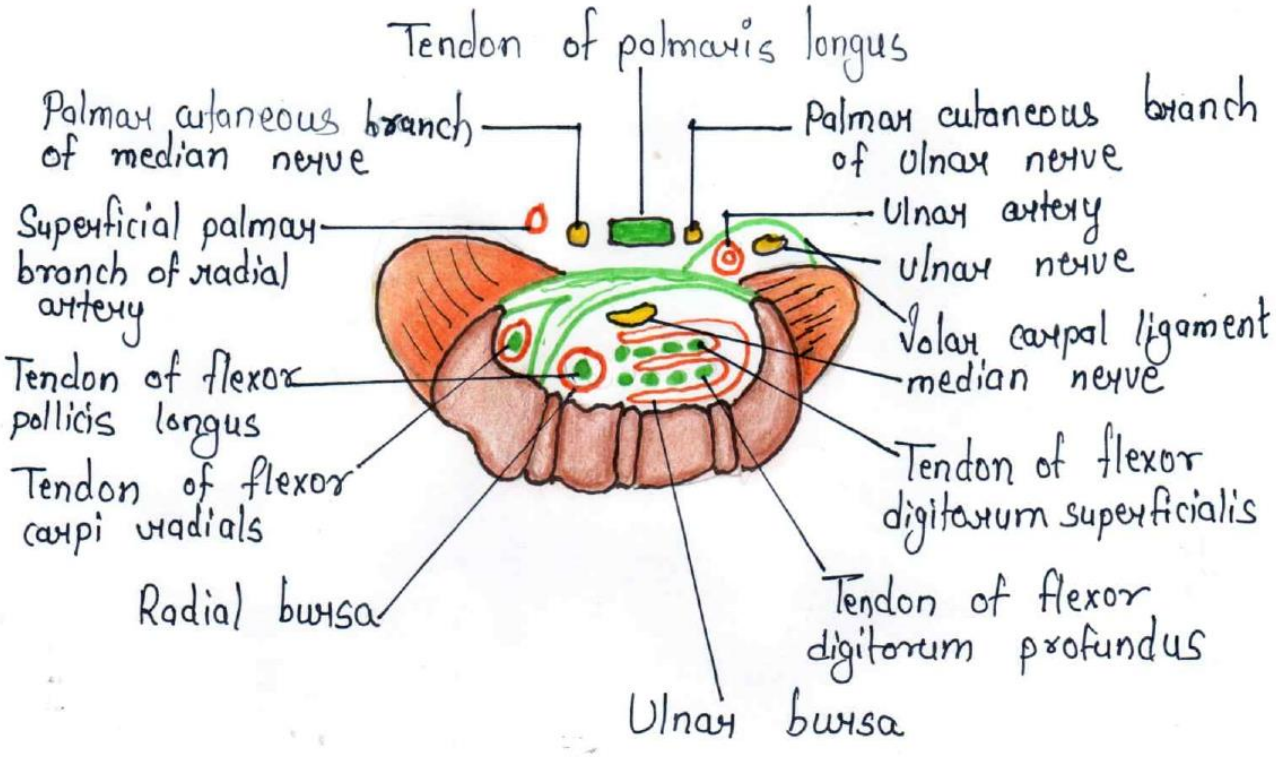
RELATION-

(A) STRUCTURE PASSES SUPERFICIAL TO FLEXOR RETINACULUM

a. PALMARIS CUTANEOUS BRANCH OF MEDIAN NERVE & ULNAR NERVE.
b. PALMARIS LONGUS TENDON.
c. ULNAR NERVE AND VESSEL.

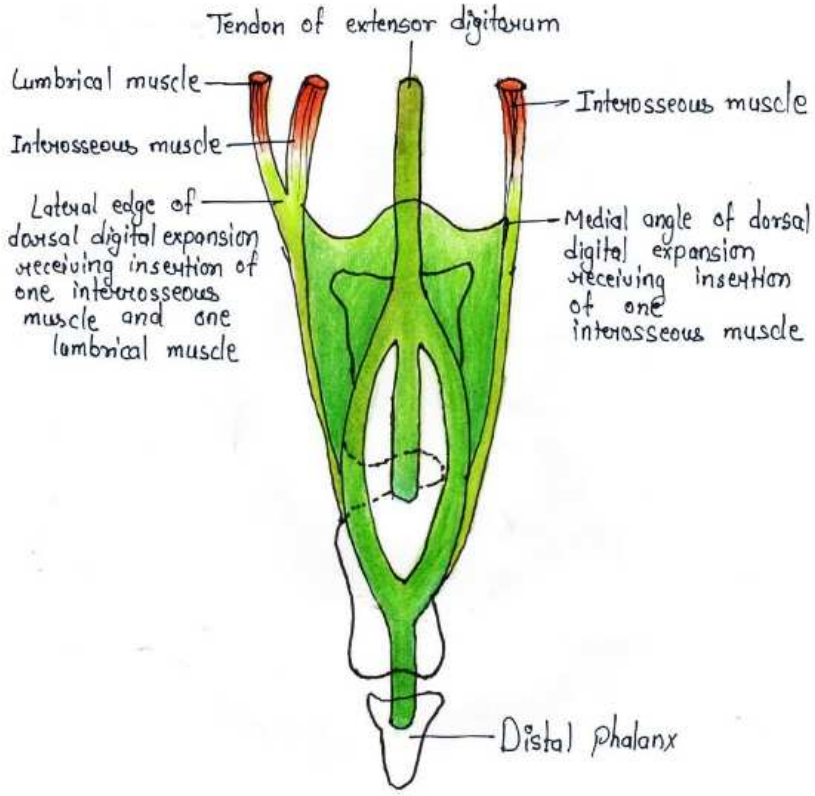
(B) STRUCTURE PASSES DEEP TO FLEXOR RETINACULUM

1) MEDIAN NERVE.
2) ULNAR & RADIAL BURSA.
3) 4 TENDON OF FLEXOR DIGITORUM SUPERFICIALIS.
4) TENDON OF FLEXOR DIGITORUM PROFUNDUS.
5) TENDON OF FLEXOR POLLICIS LONGUS.



DORSAL DIGITAL EXPANSION

- SMALL TRIANGULAR APONEUROSIS COVERING THE DORSUM OF THE PROXIMAL PHALANX.
- THE MAIN TENDON OF THE EXTENSOR DIGITORUM OCCUPIES THE CENTRAL PART OF THE EXTENSION BASE.
- THE POSTEROLATERAL CORNERS OF THE EXTENSOR EXPANSION- JOINED BY TENDONS OF THE INTEROSSEI AND OF LUMBRICAL MUSCLES.
- THE POINTS OF ATTACHMENT OF THE INTEROSSEI (PROXIMAL) AND LUMBRICAL (DISTAL) ARE OFTEN CALLED 'WING TENDONS'
- NEAR THE PROXIMAL INTERPHALANGEAL JOINT, THE EXTENSOR TENDON- DIVIDES INTO A CENTRAL SLIP AND TWO COLLATERAL SLIPS.
- THE CENTRAL SLIP IS JOINED BY SOME FIBRES FROM THE MARGINS OF THE EXPANSION AND IS INSERTED ON THE DORSUM OF THE BASE OF THE MIDDLE PHALANX.
- THE TWO COLLATERAL SLIPS ARE JOINED BY THE REMAINING THICK MARGIN OF THE EXTENSOR EXPANSION, THEN JOIN EACH OTHER AND ARE INSERTED ON THE DORSUM OF THE BASE OF THE DISTAL PHALANX.



SPACES OF HAND

ARRANGEMENT OF FASCIAE AND THE FASCIAL SEPTA IN THE HAND IN SUCH A WAY MANY SPACES ARE FORMED, SPACES ARE OF SURGICAL IMPORTANCE BECAUSE THEY MAY BECOME INFECTED- IMPORTANT SPACES ARE AS FOLLOWS :-

A. PALMAR SPACES → PULP SPACE OF FINGER, MID PALMAR SPACE, THENAR SPACE.
B. DORSAL SPACE → DORSAL SUBCUTANEOUS AND DORSAL SUBAPONEUROTIC SPACE.
C.THE FOREARM SPACE OF PARONA.

MID PALMAR SPACE

1. SHAPE → TRIANGULAR

2. SITUATION → UNDER INNER HALF OF HOLLOW OF PALM

3. EXTEND → PROXIMAL – FOREARM SPACE OF PARONA
   DISTAL – DISTAL PALMAR CREASE.

4.BOUNDARY → ANTERIOR – FLEXOR TENDONS OF 3RD,4TH,& 5TH DIGITS.
- 2ND, 3RD, & 4TH LUMBRICALS.
- PALMAR APONEUROSIS.

→ POSTERIOR – FASCIA COVERING INTEROSSEI & METACARPALS.
→ LATERAL – INTERMIDIATE PALMAR SEPTUM.
→ MEDIAL – MEDIAL PALMAR SEPTUM.

5.DRAINAGE → INCISION IN EITHER THE 3RD OR 4TH WEB SPACE.

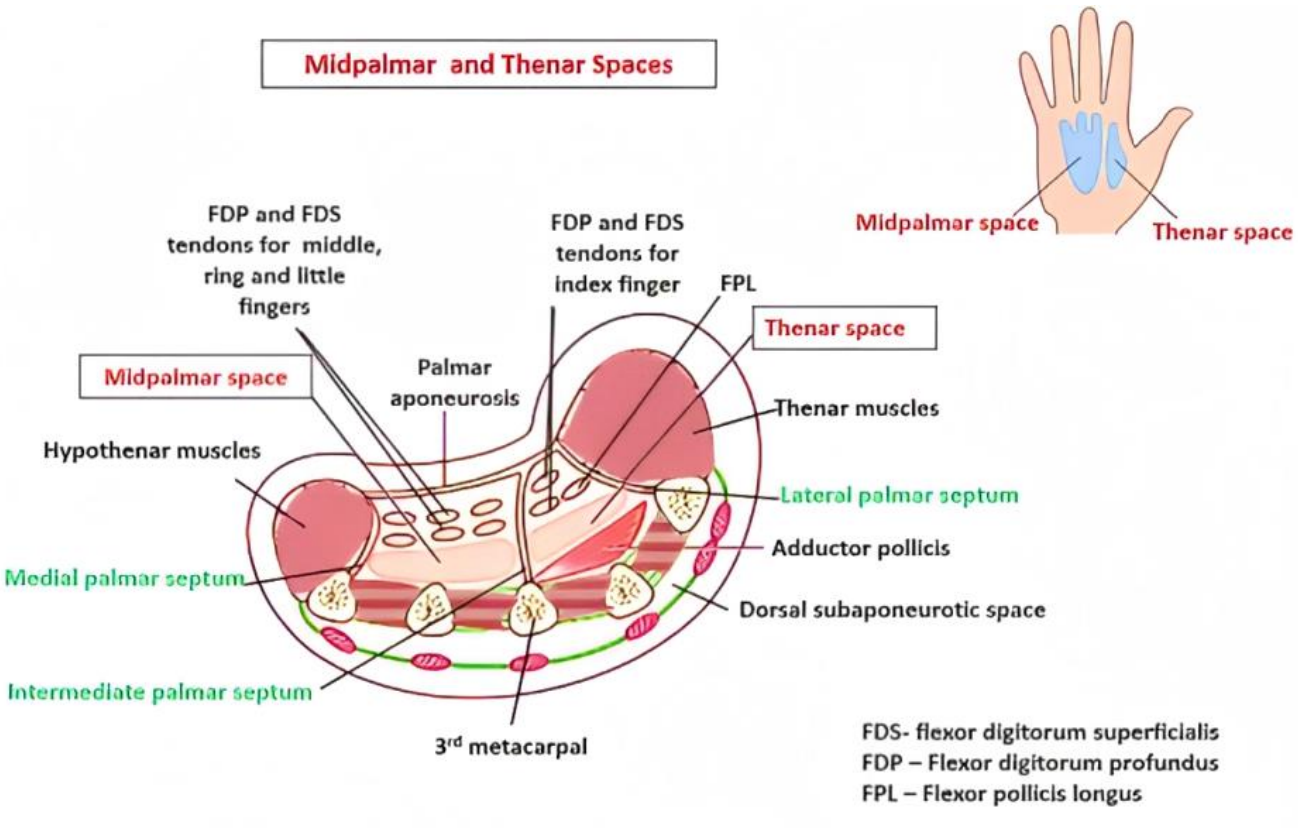
CLINICAL ANATOMY

MIDPALMAR SPACE INFECTION

- ULNAR BURSA IS CONSIDERED AS THE INLET FOR INFECTION AND LUMBRICAL CANAL AS THE OUTLET OF INFECTION IN PALMAR SPACE
- PUS FRONT SPACE -> DRAINED BY INCISION IN MEDIAL TWO WEB SPACES

PULP SPACE INFECTION

- BEING THE MOST EXPOSED PART OF DIGIT THE PULP SPACES ARE PRONE FOR INFECTION
- ABSCESS IN PULP SPACE IS CALLED WHITLOW OR FELON



SNUFF BOX

TRIANGULAR DEPRESSION → POSTERIOR LATERAL SIDE OF WRIST, SEEN WHEN THUMB IS EXTENDED

BOUNDARIES -

1. ANTEROLATERALLY – ABDUCTOR POLLICIS LONGUS, EXTENSOR POLLICIS BREVIS.
2. POSTEROMEDIALLY – EXTENSOR POLLICIS LONGUS.
3. FLOOR – SCAPHOID & TRAPEZIUM.
4. ROOF – SKIN & SUPERFICIAL FASCIA

CONTENT - 1.RADIAL ARTERY & RADIAL NERVE
2.CEPHALIC VEIN

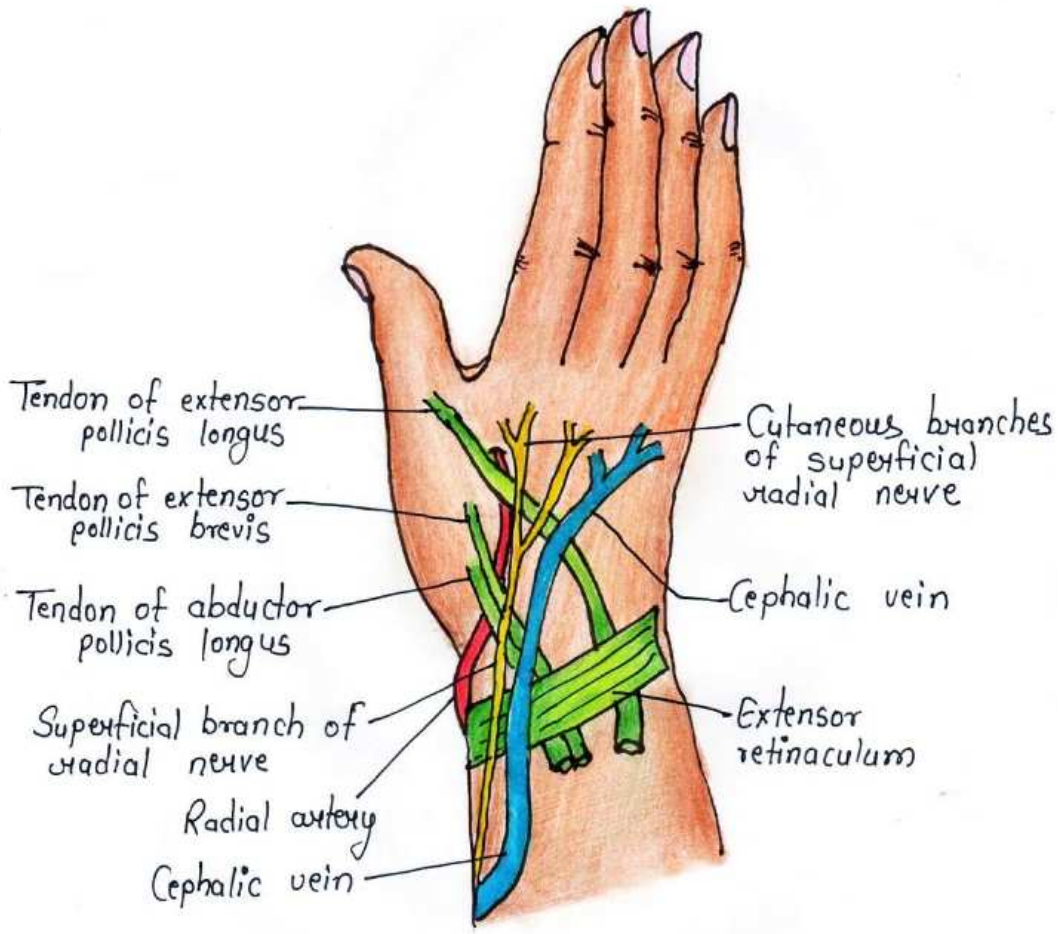
CLINICAL ANATOMY

CLINICAL SIGNIFICANCE OF ANATOMICAL SNUFF BOX

- PULSATION OF RADIAL ARTERY CAN BE FELT IN ANATOMICAL BOX
- TENDRENESS IN THE ANATOMICAL BOX INDICATES FRACTURE OF SCAPHOID BONE
- CEPHALIC VEIN AT THIS SITE IS OFTEN USED FOR GIVING INTRAVENPOUS FLUIDS

MNEMONIC FOR CONTENT

CARS (C-CEPHALIC VEIN, A – ARTERY (RADIAL), R – RADIAL NERVE, S – SUPERFICIAL BRANCH)



THENAR EMINENCE

1) ABDUCTOR POLLICIS BREVIS -

ORIGIN – TUBERCLE OF SCAPHOID, CREST OF TREPAZIUM, FLEXOR RETINACULUM
INSERTION – BASE OF PROXIMAL PHALANX OF THUMB
ACTION – ABDUCTION OF THUMB

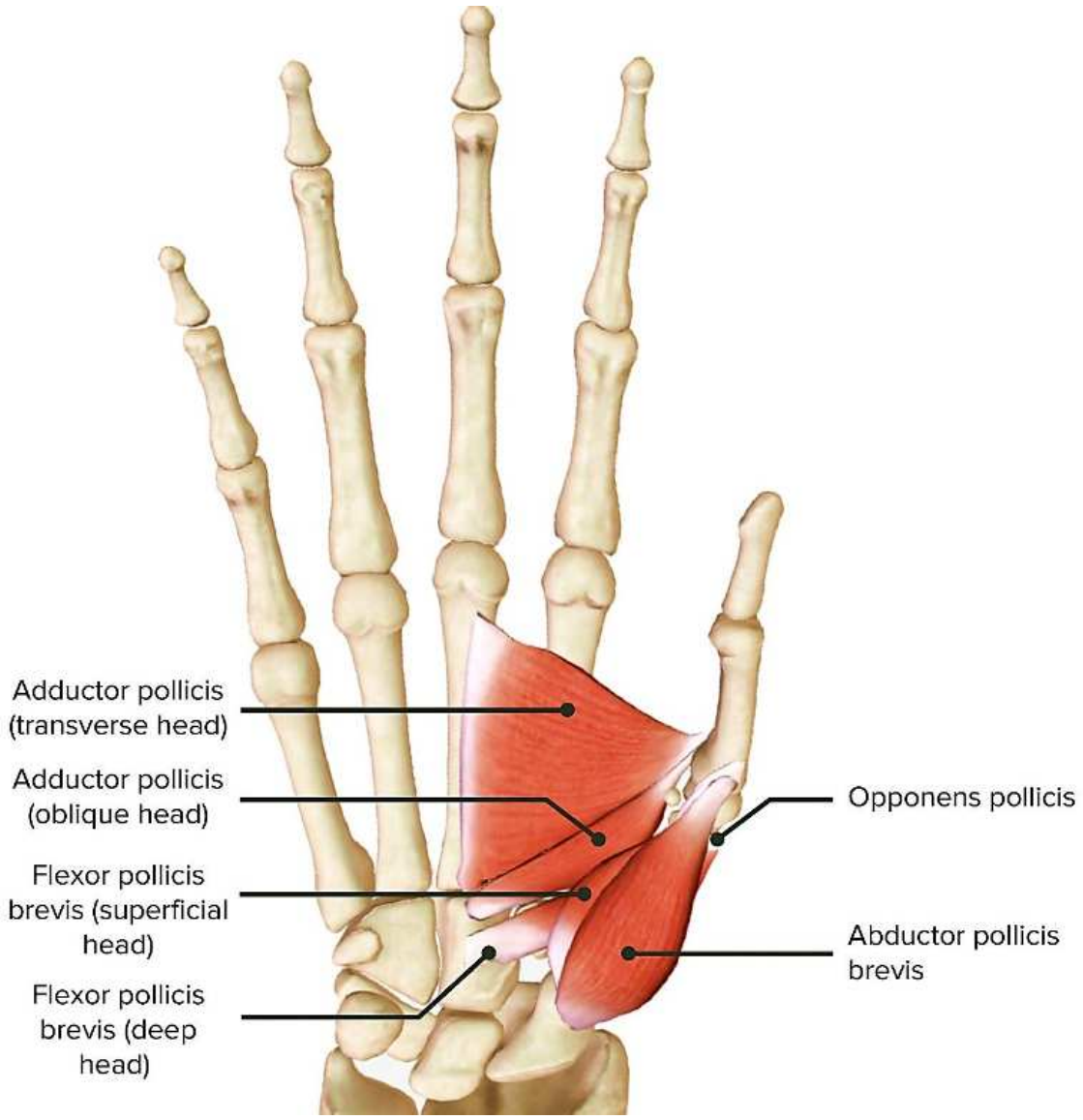
2) FLEXOR POLLICIS BREVIS -

ORIGIN – FLEXOR RETINACULUM, CREST OF TREPAZIUM & CAPITATE BONES
INSERTION - BASE OF PROXIMAL PHALANX OF THUMB
ACTION – FLEXES METACARPOPHALANGEAL JOINT OF THUMB

3) OPPONENS POLLICIS -

ORIGIN - FLEXOR RETINACULUM CREST OF TREPAZIUM
INSERTION – LATERAL HALF OF PALMAR SURFACE OF THE SHAFT OF METACARPAL BONE OF THUMB
ACTION – PULLS THUMB MEDIALLY AND FORWARD ACROSS PALM (OPPOSES THUMB TOWARDS THE FINGERS)

4) NERVE SUPPLY OF THENAR MUSCLE - MEDIAN NERVE



LUMBRICLE MUSCLES

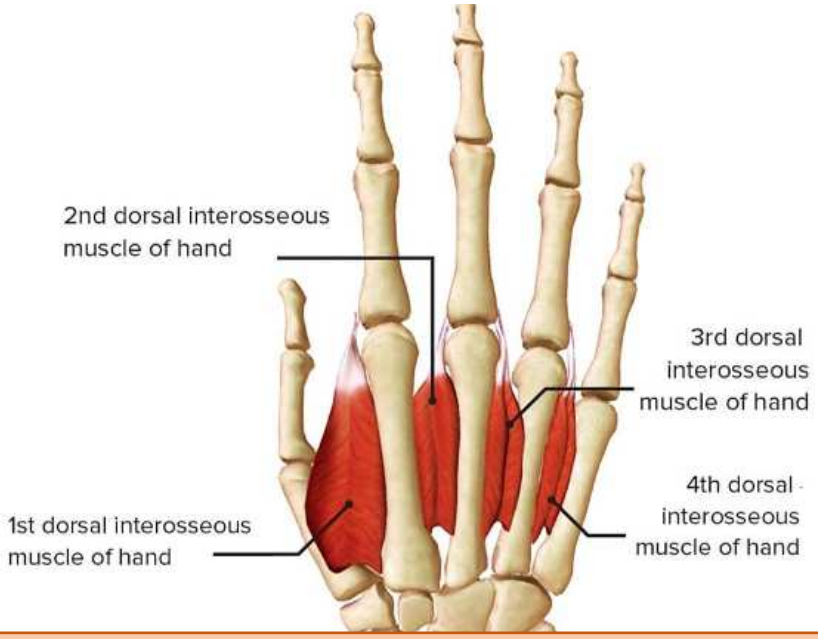
ORIGIN -

1ST – LATERAL SIDE OF TENDON OF FLEXOR DIGITORUM PROFUNDUS OF 2ND DIGIT.
2ND – LATERAL SIDE OF SAME TENDON OF 3RD DIGIT.
3RD – ADJACENT SIDES OF SAME TENDONS OF 3RD & 4TH DIGIT.
4TH – ADJACENT SIDES OF SAME TENDONS OF 4TH & 5TH DIGIT.

INSERTION –
EXTENSOR EXPANSION → DORSUM OF BASE OF → DISTAL PHALANGES

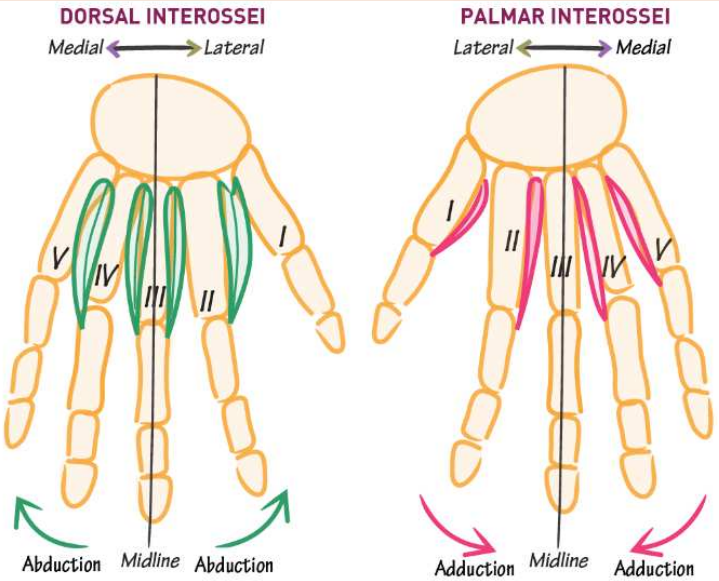
NERVE SUPPLY – 1ST & 2ND → MEDIAN NERVE
– 3RD & 4TH → ULNAR NERVE

ACTION – FLEX METACARPOPHALANGEAL JOINTS
– EXTEND INTERPHALANGEAL JOINT



IMPORTANT INFORMATION

- ALL DORSAL INTEROSSEI CAUSE ABDUCTION OF THE DIGITS AWAY FROM THE LINE OF THE MIDDLE FINGER. THE FIRST AND FIFTH DIGITS DO NOT REQUIRE DORSAL INTEROSSEI AS THEY HAVE THEIR OWN ABDUCTORS.
- BOTH PALMAR AND DORSAL INTEROSSEI FLEX THE METACARPOPHALANGEAL JOINTS AND EXTEND THE INTERPHALANGEAL JOINTS.
- PALMAR INTEROSSEI ADDUCT FINGERS TOWARDS CENTRE OF THIRD DIGIT OR MIDDLE FINGER



SHOULDER JOINT

TYPE - SYNOVIAL JOINT OF BALL & SOCKET VARIETY.

ARTICULATING SURFACE - GLENOID CAVITY OF SCAPULA & HEAD OF HUMERUS.

WEAK JOINT BECAUSE GLENOID CAVITY IS SMALL & SHALLOW
STABILITY OF JOINT IS MAINTAIN BY → CAROCOACROMIAL ARCH
→ ROTATOR CUFF
→ GLENOID LABRUM (LIP.)

LIGAMENT

1.CAPSULAR LIGAMENT- ATTACHMENT- VERY LOOSE AND PERMIT FREE MOVEMENT
MEDIALLY- SUPRAGLENOID TENDON
LATERAL- ANATOMICAL NECK OF HUMERUS
ANTERIOR- CAPSULE IS REINFORCED BY SUPERIOR,
MIDDLE, INFERIOR GLENOHUMERAL LIGAMENT

2.CORACOHUMERAL LIGAMENTS - STRONG BAND OF FIBROUS TISSUE THAT PASES FROM BASE OF CORACOID PROCESS TO ANTERIOR ASPECT OF GREATER TUBERCLE OF HUMERUS

3.TRANSVERSE HUMERAL LIGAMENT - BRIDGE UPPER PART OF BICIPITAL GROOVE OF HUMERUS.
-TENDON OF LONG HEAD OF BICEP BRACHII PASS DEEP TO THE LIGAMENT

4.GLENOID LABRUM - FIBRO CARTILAGINOUS RIM COVERS GLENOID CAVITY

5.ACCESSORY LIGAMNETS
CORACOACROMIAL LIGAMENT - ATTACHMENT - ROOT OF CORACOID PROCESS TO NECK OF HUMERUS.

BURSAE- SUBACROMIAL BURSA, SUBSCAPULARIS BURSA, INFRASPINATOUS BURSA

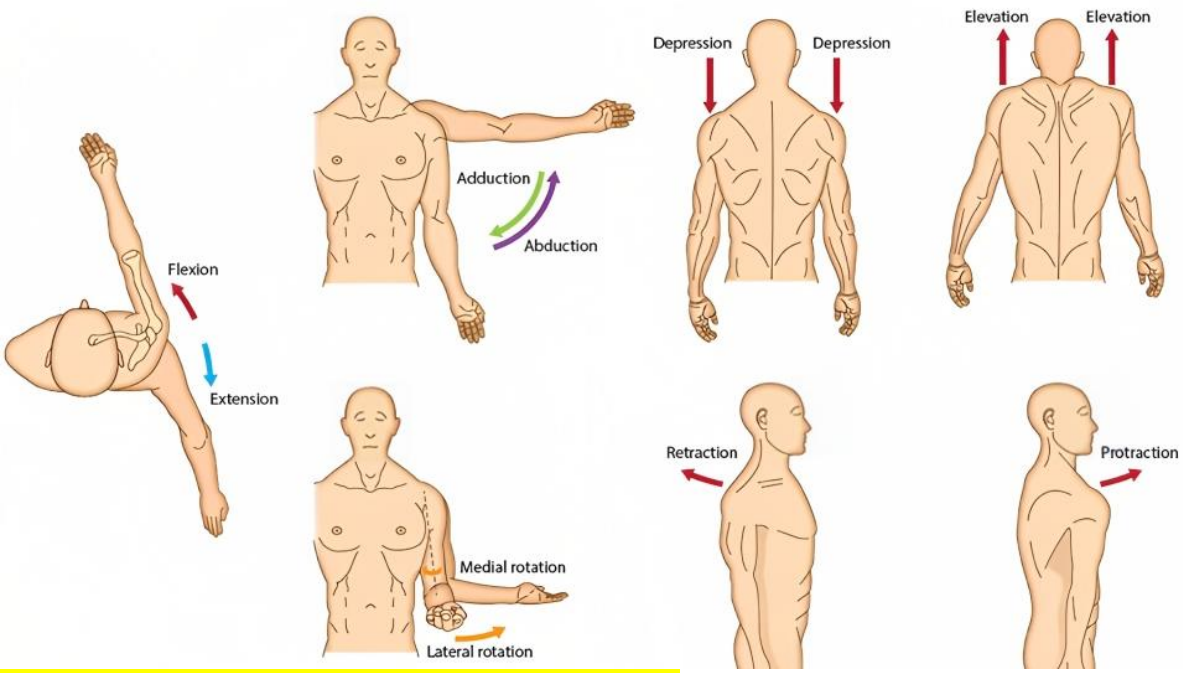
RELATION - SUPERIOR - CORACOACROMIAL ARCH, SUBACROMIAL BURSA
-SUPRASPINATOUS & DELTOID
INFERIOR - LONG HEAD OF TRICEP BRACHII
- AXILLARY NERVE
- POSTERIOR CIRCUMFLEX HUMERAL ARTERY
ANTERIOR - SUBSCAPULARIS
- CORACOBRACHIALIS & SHORT HEAD OF BICEP BRACHII
POSTERIOR - INFRASPINATOUS & TERES MINOR

BLOOD SUPPLY - ANT. & POST. CIRCUMFLEX HUMERAL ARTERY
- SUPRASCAPULAR VESSEL & SUBSCAPULAR VESSEL

NERVE SUPPLY - AXILLARY NERVE SUPPLY
- MUSCULOCUTANOUS NERVE
- SUPRASCAPULAR NERVE

MOVEMENT OF SHOULDER JOINT

1. FLEXION – ARM MOVE FORWARD & MEDIAL
2. EXTENSION – ARM MOVE LATERAL BACKWARD
3. ADDUCTION – OPPOSITE OF ABDUCTION
4. ABDUCTION – ARM MOVE ANTEROLATERALLY AWAY FROM TRUNK
5. MEDIAL ROTATION – HAND MOVE MEDIAL ACROSS THE CHEST
6. LATERAL ROTATION – HAND MOVE LATERAL OF THE SHOULDER JOINT
7. CIRCUMDUCTION – COMBINATION OF DIFFERENT MOVEMENT



FACTORS PROVIDING STABILITY TO SHOULDER JOINT

1) ROTATOR CUFF
2) CORACOACROMIAL ARCH
3) LONG HEAD OF BICEP TENDON
4) GLENOID LABRUM

CLINICAL ANATOMY

DISLOCATION OF SHOULDER JOINT

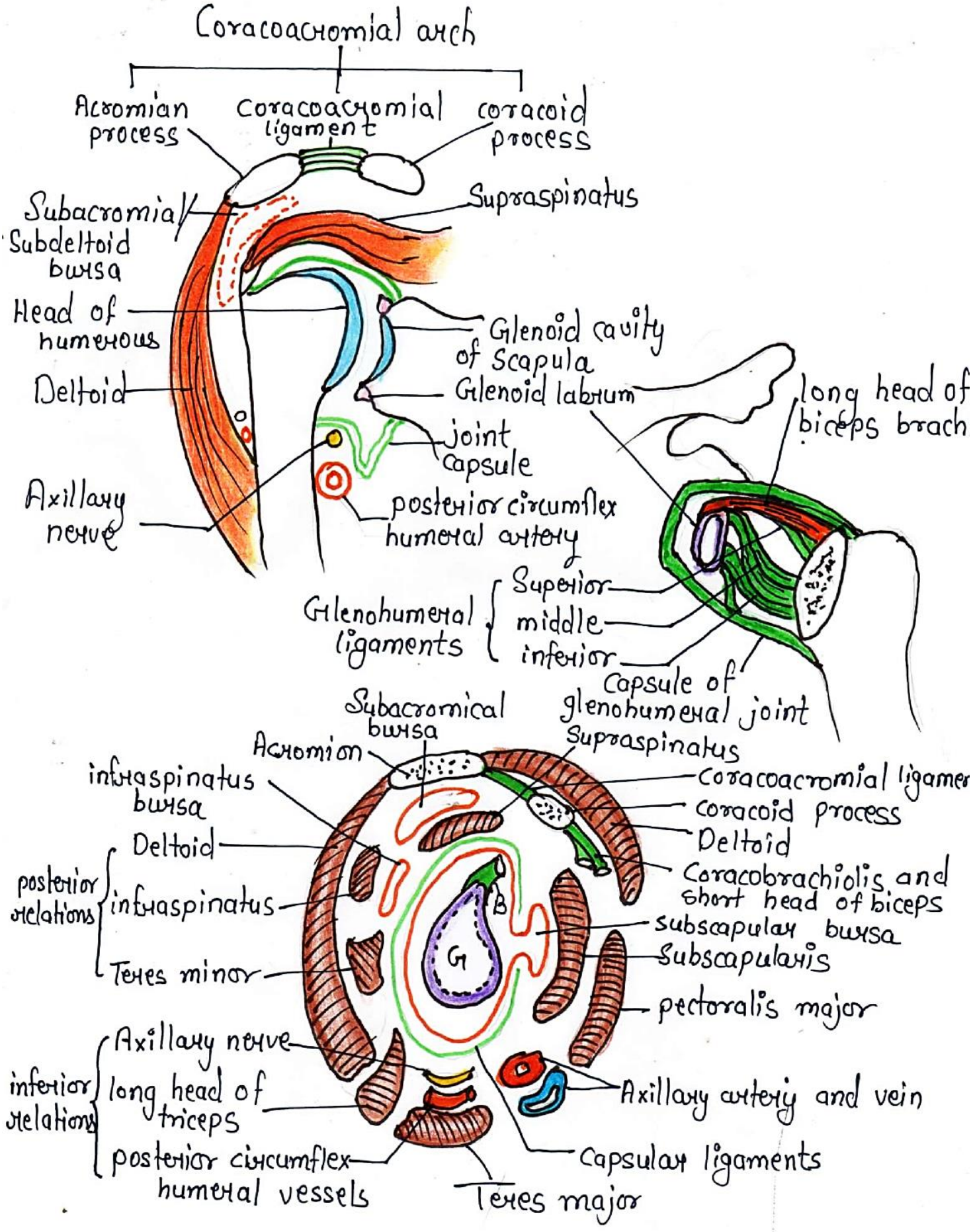
- DISLOCATION OF JOINT MOSTLY OCCUR INFERIORLY BECAUSE THE JOINT IS LEAST SUPPORTED ON THIS ASPECT
- DISLOCATION IS UASUALLY CAUSED BY EXCESSIVE EXTENSION AND LATERAL ROTATION OF HUMERUS
- CLINICALLY, IT PRESENT AS :
  - HOLLOW IN ROUNDED CONTOUR OF SHOULDER
  - PROMINENCE OF SHOULDER TIP

FROZEN SHOULDER (ADHESIVE CAPSULITIS)

CLINICAL CONDITION CHARACTERIZED BY PAIN AND UNIFORM LIMITATION OF ALL MOVEMENTS OF SHOULDER JOINT, WITH NO RADIOLOGICAL CHANGE IN JOINT
OCCURS -> DUE TO SHRINKAGE OF JOINT CAPSULE,HENCE THE NAME ADHESIVE CAPSULITIS
SEEN -> 40-60 YEARS OF AGE

ROTATOR CUFF DISORDERS

MOST COMMON DISORDER OF ROTATOR CUFF INCLUDE CALCIFIC SUPRASPINATUS TENDINITIS AND SUBACROMIAL BURSITIS


Coracoachromial arch
Acromian process
Coracoachromial ligament
coracoid process
Supraspinatus
Subacromial
Subdeltoid bursa
Head of humerous
Glenoid cavity of Scapula
Glenoid labrum
Deltoid
joint capsule
Axillary nerve
posterior circumflex humeral artery
long head of biceps brach
Glenohumeral ligaments
Superior
middle
inferior
Capsule of glenohumeral joint
Subacromical bursa
Acromion
Supraspinatus
infraspinatus bursa
Coracoacromial ligamer
coracoid process
Deltoid
posterior relations
Deltoid
infraspinatus
Teres minor
Coracobrachiolis and short head of biceps
B
G
subscapular bursa
Subscapularis
pectoralis major
inferior relations
Axillary nerve
long head of triceps
posterior circumflex humeral vessels
Axillary artery and vein
Capsular ligaments
Teres major

ELBOW JOINT

TYPE - SYNOVIAL JOINT OF VARIETY OF HINGE JOINT BETWEEN LOWER END OF HUMEROUS & UPPER END OF RADIUS & ULNA.

ARTICULATION SURFACE -

1) UPPER -> CAPITULUM & TROCLEAR CONDYLE OF HUMERUS
-> CORONOID FOSSA JUST LIE ABOVE TROCHLEAR
(CORONOID FOSSA FIT CORONOID PROCESS OF ULNA IN FLEXION)
-> RADIAL FOSSA JUST ABOVE CAPITULUM
(RADIAL FOSSA FIT RADIAL HEAD OF FLEXION)

2) LOWER → HEAD OF RADIUS WITH CAPITULUM
-TROCHLEAR NOTCH OF ULNA WITH TROCHLEAR CONDYLE OF HUMERUS

LIGAMENT

1.CAPSULAR LIGAMENT-ATTACHMENT- SUPERIOR - LOWER END OF HUMERUS
INFEROMEDIAL- MARGIN OF TROCHLEAR NOTCH OF ULNA
INFERIOLATERAL- ANGULAR LIGAMENT

2.ULNAR COLLATERAL LIGAMENT- TRIANGULAR IN SHAPE
- APEX-MEDIAL EPICHONDYLE OF HUMERUS
- BASE-ULNA
- ANTERIOR BAND- BELOW CORONOID PROCESS
- POSTERIOR BAND- BELOW OLECRANON PROCESS

3.RADIAL COLLATERAL LIGAMENT - FAN SHAPE
- EXTEND LATERAL EPICHONDYLE-→ANGULAR LIGAMENT

RELATIONS - ANTERIORLY - BRACHIALIS
- MEDIAN NERVE & BRACHIAL ARYTERY
- TENDON OF BICEPS BRACHII
POSTERIORLY - TENDON OF TRICEP & ANCONEUS
MEDIALLY - FLEXOR CARPI ULNARIS
- ULNAR NERVE
- COMMON FLEXOR ORIGIN OF MUSCLE OF FOREARM
LATERALLY - SUPINATOR
- COMMON EXTENSOR ORIGIN OF MUSCLE OF FOREARM MUSCLE
- EXTENSOR CARPI RADIALIS BREVIS

BURSA - SUBTENDINOUS OLECRANON BURSA
- SUBCUTANEOUOS OLECRANON BURSA
- BICIPITORADIAL BURSA
- BURSA SEPARATING THE BICEPS TENDON FROM THE OBLIQUE CORD

STABILITY OF JOINT - PULLEY SHAPED TROCHLEA OF HUMERUS FITS PROPERLY INTO JAW LIKE TROCHLEAR NOTCH OF ULNA
- STRONG ULNAR AND RADIAL COLLATERAL LIGAMENTS

BLOOD SUPPLY – FROM ANATOMOSES AROUND ELBOW JOINT

NERVE SUPPLY– ULNAR, MEDIAN, RADIAL & MUSCULOCUTANEOUS NERVE

MOVEMENT – FLEXION → BRACHIALIS
→ BICEP BRACHII & BRACHIORADIALIS

EXTENSION → TRICEPS BRACHII
→ ANCHONEUS

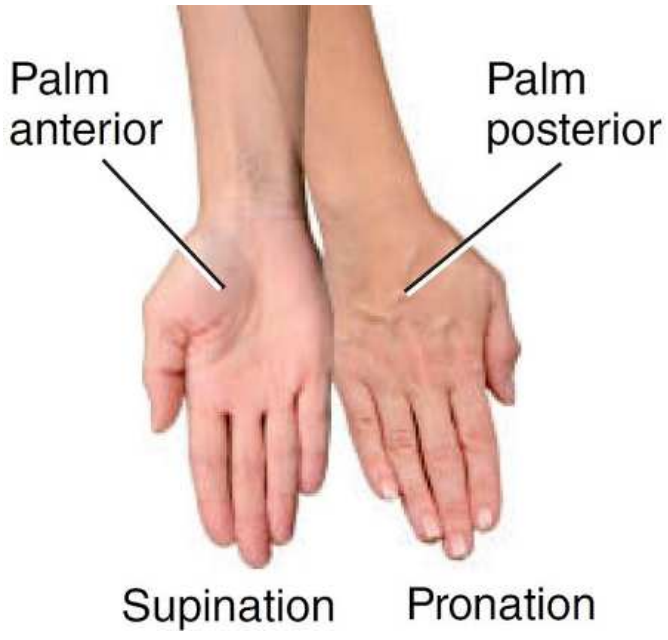
SUPINATION & PRONATION

1.SUPINATION & PRONATION ARE ROTATORY MOVEMENT OF FOREARM →VERTICAL AXIS IN SEMIFLEX ELBOW.

2.SUPINATION → PALM → TURN UPWARD.
3.PRONATION → PALM TURN DOWNWARDS.

4.VERTICAL AXIS MOVEMENT PASS CENTER OF HEAD OF RADIUS ABOVE.
5.THIS AXIS IS NOT STATIONARY BECAUSE -->LOWER END OF ULNA IS NOT FIXED.
6.IT MOVE → BACKWARD LATERALLY IN PRONATION
→ FORWARD MEDIALLY IN SUPINATION
7.SUPINATION IS POWERFUL BECAUSE IT IS ANTI GRAVITATIONAL
8.SUPINATION MOVEMENT IS RESPONSIBLE FOR ALL SCREWING MOVEMENT OF HAND

9.AROUND 50 DEGREE OF SUPINATION AND PRONATION ARE GENERALLY TO PERFORM MANY ROUTINE ACTIVITIES
10.PRONATION → CHEIFLY BY PRONATOR QUADRATUS
→ RAPIDLY MOVEMENT BY PRONATOR TERES
11.SUPINATION → CAUSE BY SUPINATOR MUSCLE & BICEP BRACHII
→ SLOW SUPINATION → BY SUPINATR MUSCLE
→ RAPID SUPINATION →BY BICEP BRACHII



CLINICAL ANATOMY

DISLOCATION OF ELBOW JOINT

- POSTERIOR DISLOCATION OF ELBOW ARE MORE COMMON ARE OFTEN ASSSOCIATED WITH FRACTURE OF CORONOID PROCESS

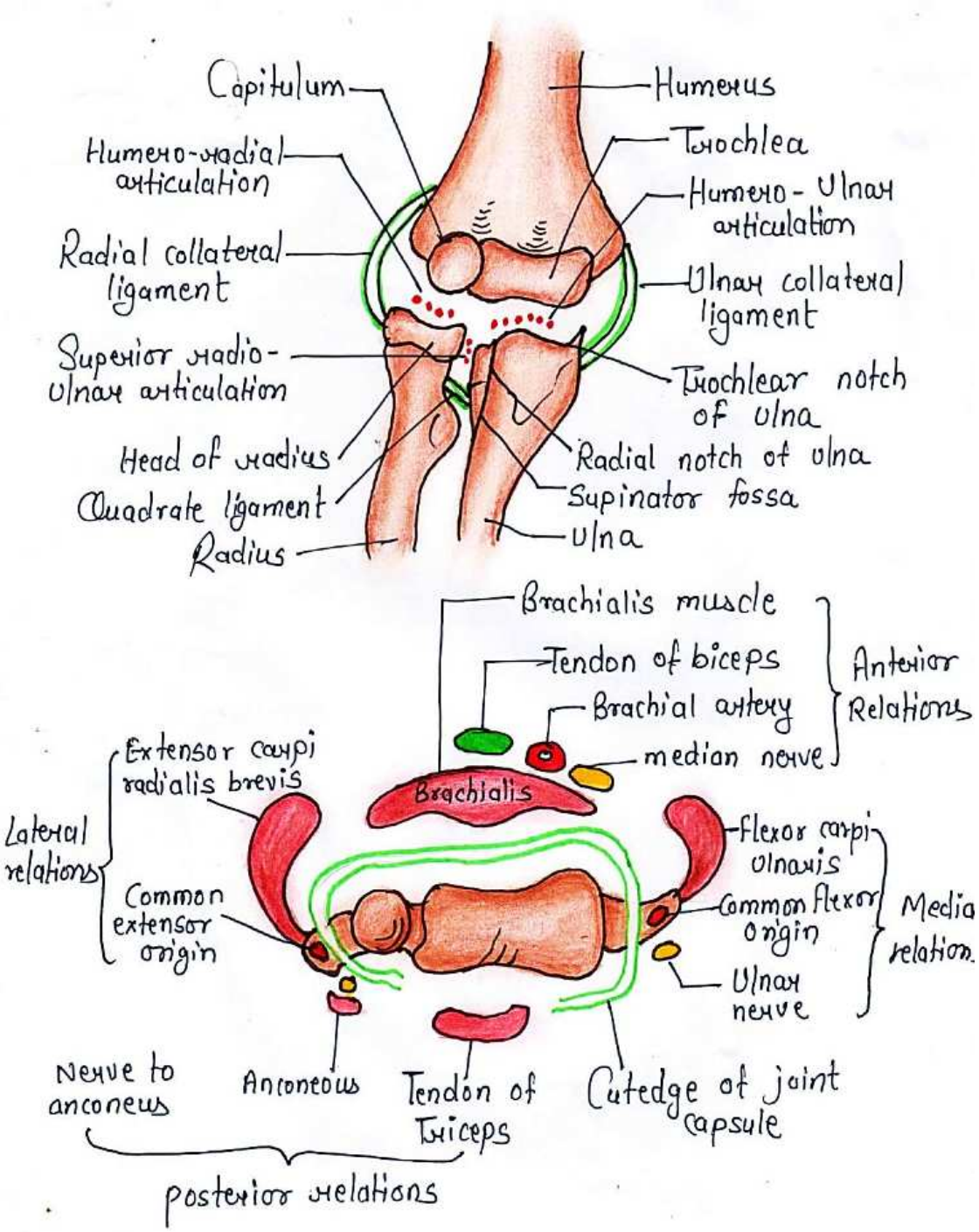
ELBOW EFFUSION

- DISTENISON OF ELBOW JOINT DUE TO EFFUSION WITHIN CAVITY OCCURS POSTERIORLY BECAUSE CAPSULE OF JOINT IS THIN POSTERIORLY AND COVERING FASCIA IS ALSO THIN
- THE JOINT IS ASPIRATED BY INSERTING A NEEDLE ON THE POSTEROLATERAL SIDE, ABOVE THE HEAD OF RADIUS WITH ELBOW AT THE RIGHT ANGLE

TENNIS ELBOW

CLINICAL CONDITION CHARACTERIZED BY PAIN AND TENDERNESS OVER THE LATERAL EPICONDYLE OF HUMERUS WITH PAIN DURING ABRUPTION PRONATION

IT OCCUR DUE TO – SPRAIN OF LATERAL COLLATERAL LIGAMENT OF ELBOW JOINT
- TEAR OF FIBRES OF EXTENSOR CARPI RADIALIS BREVIS
- STRAIN OR TEAR OF COMMON EXTENSOR ORIGIN



WRIST JOINT

THE JOINT BETWEEN FOREARM AND HAND IS CALLED WRIST JOINT

TYPE -> SYNOVIAL JOINT OF ELLIPSOID VARIETY BETWEEN LOWER END OF RADIUS AND CARPUS

THE PISIFORM DOES NOT PLAY A ROLE IN THE RADIOCARPAL ARTICULATION

ARTICULAR SURFACES UPPER -> INFERIOR SURFACE OF THE LOWER END OF THE RADIUS
-> ARTICULAR DISC OF THE INFERIOR RADIOULNAR JOINT
LOWER -> SCAPHOID, LUNATE & TRIQUETRAL BONES

LIGAMENTS

- THE ARTICULAR CAPSULE: IT IS ATTACHED TO :
  ABOVE: LOWER ENDS OF THE RADIUS AND ULNA
  BELOW: PROXIMAL ROW OF CARPAL BONES
- PALMAR RADIOCARPAL LIGAMENT:
  BEGIN: FROM ANTERIOR MARGIN OF THE LOWER END OF THE RADIUS.
  ATTACHED: BELOW TO THE ANTERIOR SURFACES OF THE SCAPHOID, THE LUNATE AND TRIQUETRAL BONES.
- PALMAR ULNOCARPAL LIGAMENT:
  BEGIN: ABOVE FROM THE BASE OF THE STYLOID PROCESS OF THE ULNA
  ATTACHED: TO THE LUNATE AND TRIQUETRAL BONES.
- DORSAL RADIOCARPAL LIGAMENT:
  BEGIN: ABOVE FROM THE POSTERIOR MARGIN OF THE LOWER END OF THE RADIUS
  ATTACHED: TO THE DORSAL SURFACES OF THE SCAPHOID, LUNATE AND TRIQUETRAL BONES
- RADIAL COLLATERAL LIGAMENT:
  EXTENDS FROM THE TIP OF THE STYLOID PROCESS OF THE RADIUS TO THE LATERAL SIDE OF THE SCAPHOID BONE.
- ULNAR COLLATERAL LIGAMENT:
  EXTENDS FROM- TIP OF THE STYLOID PROCESS OF THE ULNA TO THE TRIQUETRAL AND PISIFORM BONES.

RELATIONS

- ANTERIOR: LONG FLEXOR TENDONS WITH THEIR SYNOVIAL SHEATHS, AND MEDIAN NERVE
- POSTERIOR: EXTENSOR TENDONS OF THE WRIST AND FINGERS WITH THEIR SYNOVIAL SHEATHS.
- LATERAL: RADIAL ARTERY

BLOOD SUPPLY: ANTERIOR AND POSTERIOR CARPAL ARCHES.

NERVE SUPPLY: ANTERIOR AND POSTERIOR INTEROSSEOUS NERVES.

MOVEMENT: ACCOMPANIED BY MOVEMENTS AT THE MIDCARPAL JOINT

1. FLEXION: MORE AT THE MIDCARPAL THAN AT THE WRIST JOINT. THE MAIN FLEXORS ARE:

i. FLEXOR CARPI RADIALIS
ii. FLEXOR CARPI ULNARIS

iii. PALMARIS LONGUS

2. EXTENSION: IT TAKES PLACE MAINLY AT THE WRIST JOINT. THE MAIN EXTENSORS ARE:
i. EXTENSOR CARPI RADIALIS LONGUS
ii. EXTENSOR CARPI RADIALIS BREVIS
iii. EXTENSOR CARPI ULNARIS.

3. ABDUCTION (RADIAL DEVIATION): IT OCCURS MAINLY AT THE MIDCARPAL JOINT. THE MAIN ABDUCTORS ARE:
i. FLEXOR CARPI RADIALIS
ii. EXTENSOR CARPI RADIALIS LONGUS AND EXTENSOR CARPI RADIALIS BREVIS
iii. ABDUCTOR POLLICIS LONGUS AND EXTENSOR POLLICIS BREVIS

4. ADDUCTION (ULNAR DEVIATION): IT OCCURS MAINLY AT THE WRIST JOINT. THE MAIN ADDUCTORS ARE:
i. FLEXOR CARPI ULNARIS
ii. EXTENSOR CARPI ULNARIS

5. CIRCUMDUCTION: THE RANGE OF FLEXION IS MORE THAN THAT OF EXTENSION.

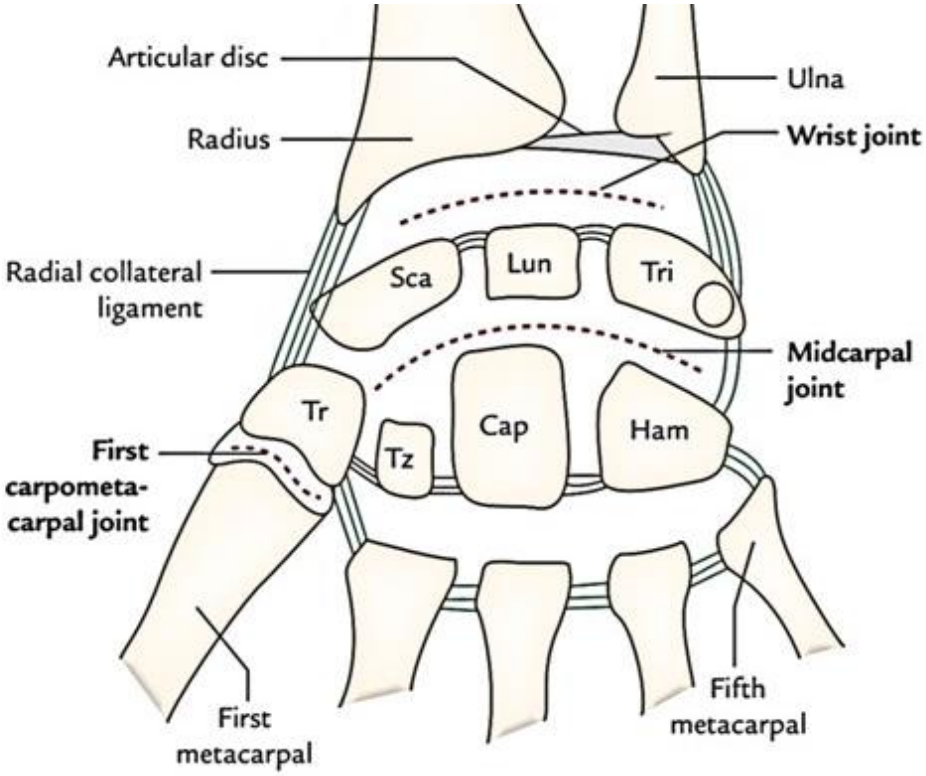
CLINICAL ANATOMY

IMMOBILIZATION OF WRIST JOINT

THE WRIST JOINT IS IMMOBILIZED IN ITS OPTIMUM POSITION OF 30 DEGREE DORSIFLEXION

GANGLION

NON TENDER CYSTIC SWELLING, WHICH SOMETIMES APPEAR ON WRIST MOST COMMONLY ON ITS DORSAL ASPECT
SIZE VARIES FROM A SMALL GRAPE TO PUL,
ITS OCCUR DUE TO MUCOID DEGENRATION OF SYNOVIAL SHEATH AROUND TENDON



PREVIOUS YEAR UNIVERSITY QUESTIONS

LONG QUESTIONS - 20MARKS

1.DESCRIBE BRACHIAL PLEXUS WITH ITS APPLIED ANATOMY ?



STUDY TIPS FOR GOOD MARKS

1) IN CLINICAL ANATOMY OR APPLIED ANATOMY - YOU HAVE TO WRITE ONLY 3-4 CLINICAL ANATOMY POINTS
2) IF IT IS MENTION TO WRITE ON SPECIFIC CLINICAL ANATOMY

FOR EXAMPLE -> WRITE CLINICAL ANATOMY OF ERBS PALSY THEN YOU HAVE TO WRITE ONLY ABOUT ERB PALSY - IN 1-1/2 PAGE.

2.DESCRIBE MEDIAL NERVE UNDER FOLLOWING HEADS -
1.ROOT VALUE 2.COURSE & RELATION 3.DISTRIBUTION 4.CLINICAL ANATOMY

HOW TO WRITE IN UNIVERSITY EXAM (TIME MANAGEMENT)

1.IN MULTIPLE HEADING QUESTION -> SPECIFIC MARKS ARE GIVEN TO EACH HEADING

FOR EXAMPLE - (2)MARKS FOR ROOT VALUE, (8)MARKS FOR COURSE, (5)MARKS FOR DISTRIBUTION & (5)MARKS FOR CLINICAL ANATOMY.

2.MARKS DISTRIBUTION OF EACH HEADING ARE GIVEN IN PAPER.

3.WRITE ACCORDING TO MARKS DISTRIBUTION.

(ZYADA LIKHNE SE ZYADA MARKS NAHI MILENGE, 2MARKS HAIN TO 2MARKS HI MILENGE - ZYADA LIKHNE KE CHAKKAR ME TIME WASTE MAT KARNA PAPER MISS HO JAAYEGA)

3. A 30-YEAR-OLD MALE PRESENTED TO EMERGENCY DEPARTMENT AFTER A ROAD TRAFFIC ACCIDENT WITH COMPLAINING OF RIGHT ARM PAIN AND INABILITY TO MOVE HIS EXTREMITY. THE FOLLOWING MORNING, HE DEVELOPED CLAWING. NERVE CONDUCTION STUDIES ON THE PERIPHERAL NERVES OF HIS ARM CONFIRMED THE
DIAGNOSIS OF INJURY TO THE ULNAR NERVE. KEEPING THIS SCENARIO IN MIND, ANSWER THE FOLLOWING -
QUESTIONS

1) ORIGIN, ROOT VALUE OF THE NERVE 2) COURSE & DISTRIBUTION 3) BRANCHES 4) EXPLAIN REASON FOR DEVELOPING THE CLAW HAND

HOW TO SOLVE CASE STUDY !!!!!!

1.MAINLY NERVES (MEDIAN, RADIAL, ULNAR NERVE) ARE GIVEN IN CASE STUDY.

2.IDENTIFICATION TRICK -
1.IF MENTION INJURY TO SPECIFIC NERVE THEN WRITE ABOUT THAT(FOR EXAMPLE - IN ABOVE CASE STUDY IT MENTION'S ULNAR NERVE INJURY)

2.IF INJURY IS NOT MENTIONED THEN SEE SENSORY LOSS

-IN RADIAL NERVE INJURY SENSORY LOSS – IN DORSAL LATERAL HALF
-IN ULNAR NERVE INJURY SENSORY LOSS IN – MEDIAL HALF OF BOTH DORSAL & PALMAR SIDE
-IN MEDIAN NERVE INJURY SENSORY LOSS IN – LATERAL HALF OF PALMAR SIDE.

(IDENTIFY AARAM SE KARNA, GALAT CASE STUDY IDENTIFY KARNE SE SEEDHE 20MARKS KA LOSS HOGA AUR FAIL.)

4.MID PALMAR SPACE IN HAND

HOW TO DRAW A DIAGRAM !!!!!!

1.TRY TO MAKE COLOURFUL DIAGRAM – 1.RED COLOUR – ARTERY
2.BLUE – VEIN
3.BROWN – MUSCLE
4.YELLOW – NERVE
5.GREEN – LIGAMMENT



2.IN LONG QUESTION TRY TO MAKE 2-3 DIAGRAMS ONLY.
(NUMBER OF DIAGRAM = MARKS)
(ZYAADA DIAGRAM BANANE ME TIME WASTE MAT KARNA WARNA NEXT QUESTION MISS HO JAYEGA – TIME MANAGE KARE CHALNA)

3.TRY TO MAKE CLEAR, COLOURFUL & WELL LABELLED DIAGRAM.

5.DESCRIBE MAMMARY GLAND?

HOW TO WRITE IN LONG QUESTIONS !!!!!!

1.IF IN LONG QUESTIONS HEADINGS ARE NOT GIVEN, THEM WRITE IN FOLLOWING HEADINGS
1.DEFINITION
2.EXTERNAL FEATURES
3.BLOOD SUPPLY
4.NERVE SUPPLY
5.LYMPHATIC DRAINAGE
6.CLINICAL ANATOMY

6.DESCRIBE SHOULDER JOINT UNDER THE FOLLOWING HEADINGS –
1.TYPE OF JOINT & ARTICULAR SURFACE 2.CAPSULAR LIGAMMENT 3.RELATION
4.APPLIED ANATOMY

7. DESCRIBE ELBOW JOINT UNDER THE FOLLOWING HEADINGS –
1.TYPE OF JOINT & ARTICULAR SURFACE 2.LIGAMENT 3.APPLIED ANATOMY

SHORT QUESTIONS – 5MARKS

HOW TO WRITE IN SHORT QUESTIONS !!!!!!



1) IN SHORT QUESTIONS WRITE 2-3 PAGES WITH ONE DIAGRAM & 1 CLINICAL ANATOMY – MAXIMUM 2 .MINIMUM 1-2 PAGES WITH DIAGRAM.

(ZYADA LIKHNE ME TIME WASTE MAT KARNA, 5 ME SE 10 MARKS NAHI MILENGE 5 ME 5 HI MILENGE, ZYADA LIKHNE KE CHAKKAR ME PAPER AUR MISS HO JAYEGA)

IMPORTANT SHORT NOTES FOR UNIVERSITY EXAMS

1) AXILLA
2) AXILLARY ARTERY
3) CUBITAL FOSSA
4) ROTATOR CUFF
5) FLEXOR RETINACULUM
6) SNUFF BOX
7) LUMBRICAL
8) THENAR MUSCLE
9) SUPINATION & PRONATION
10) FLEXOR RETINACULUM



UPPER LIMB ANATOMY IS COMPLETED !!

'BEST WAY TO PREDICT FREE FUTURE IS TO CREATE IT'

ALL THE BEST FOR YOUR EXAMS

DR.JOHARI

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